

68320 Nanofabrication and Nanocharacterization Techniques

Lecture 3

Introduction to focused ion beam milling & electron beam cathodoluminescence analysis: destructive and non-destructive depth profiling

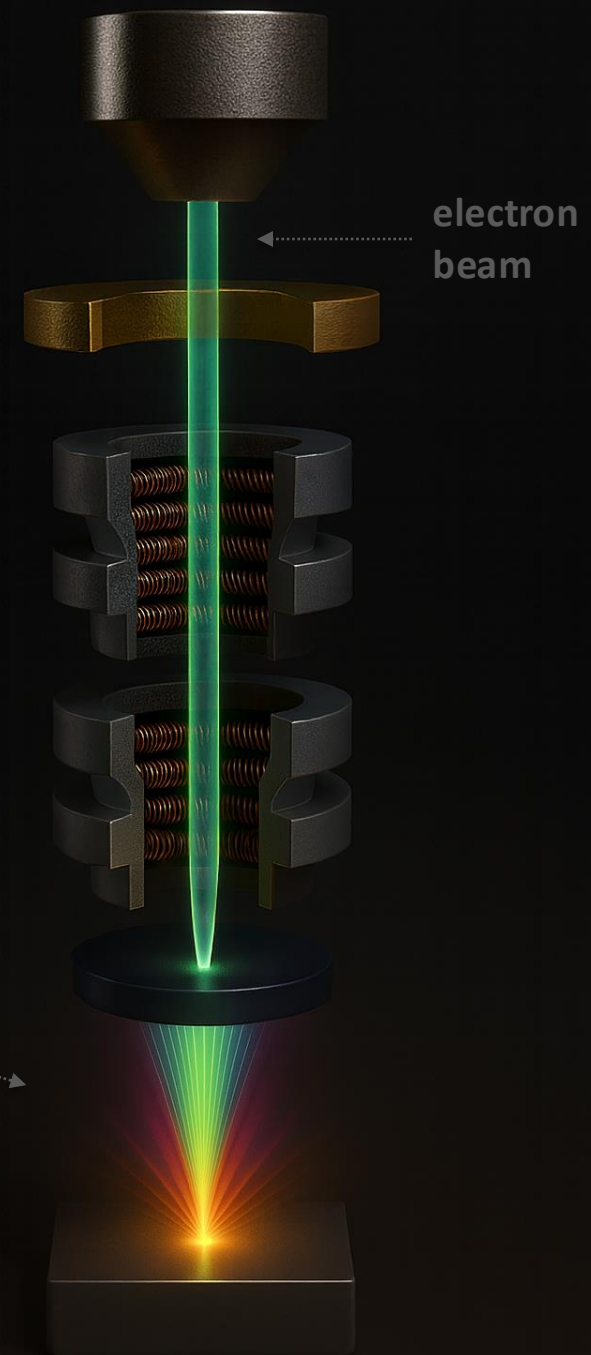
signals emitted from a sample

photons:

- cathodoluminescence
- x-rays

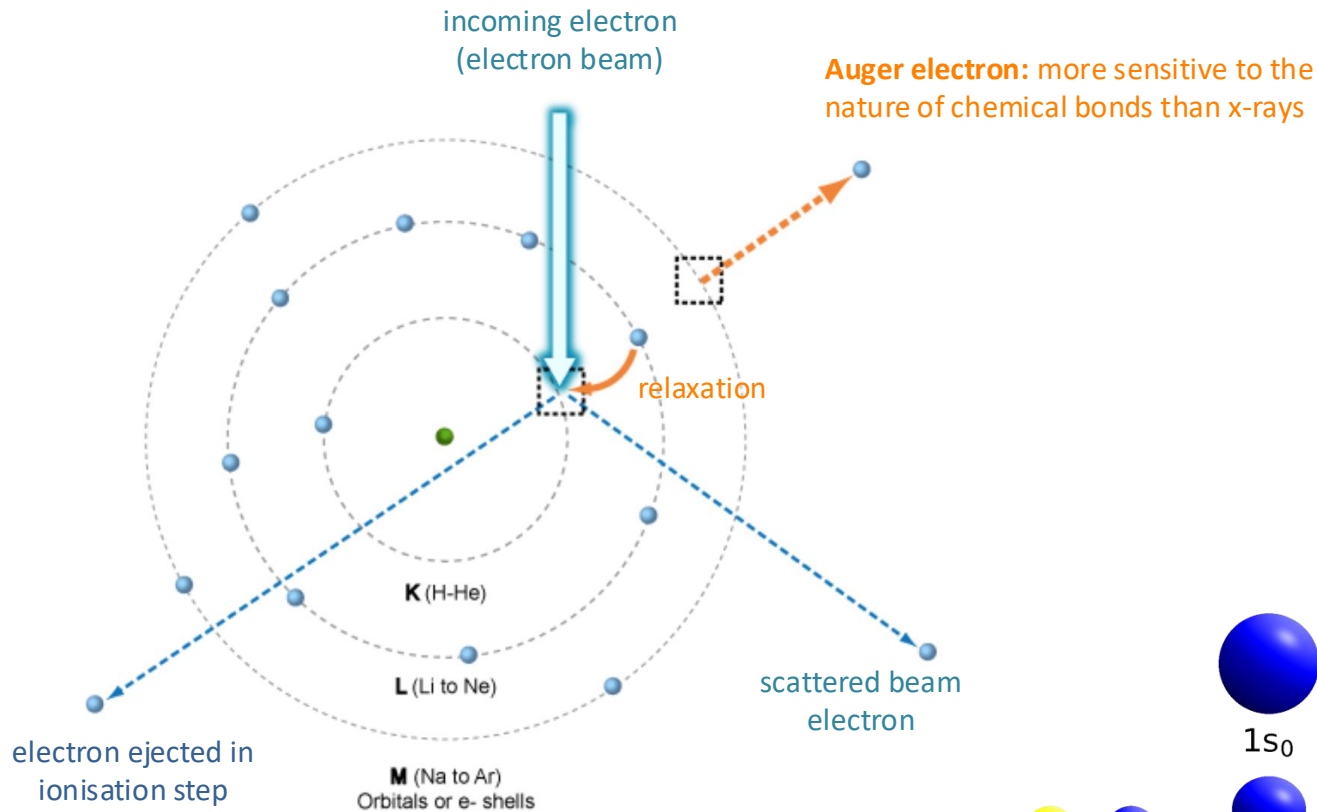
electrons:

- backscattered electrons
- secondary electrons
- Auger electrons



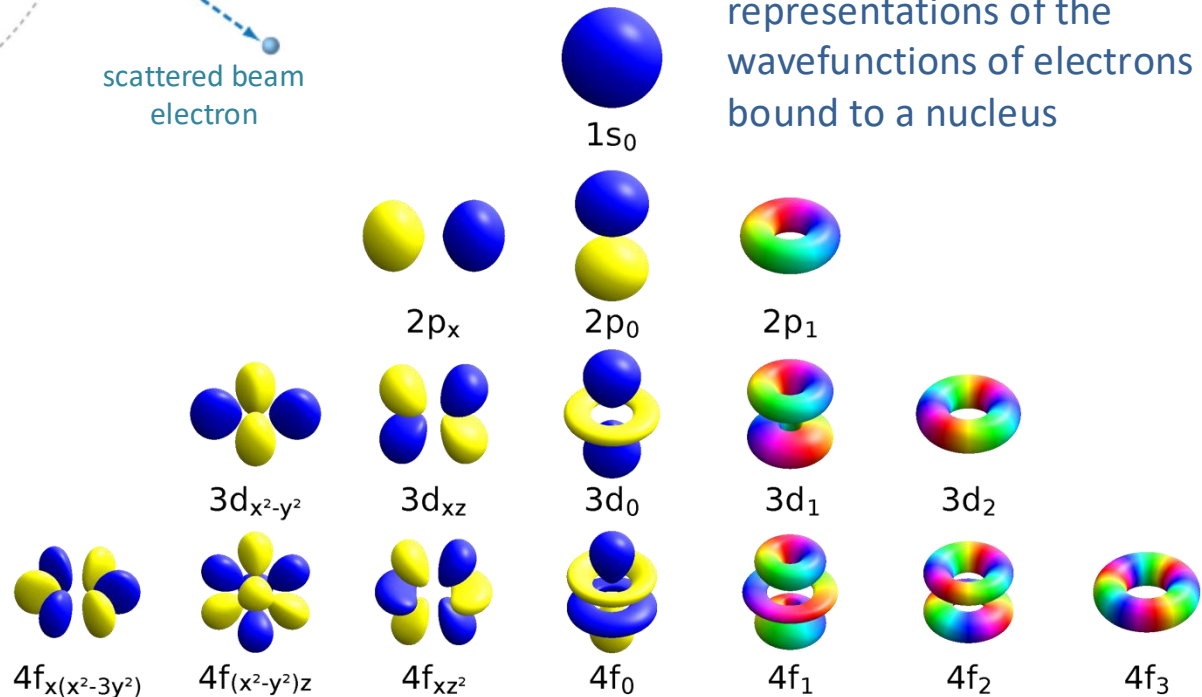
Electron and photon wavefunctions

Recap: Auger electrons & orbitals



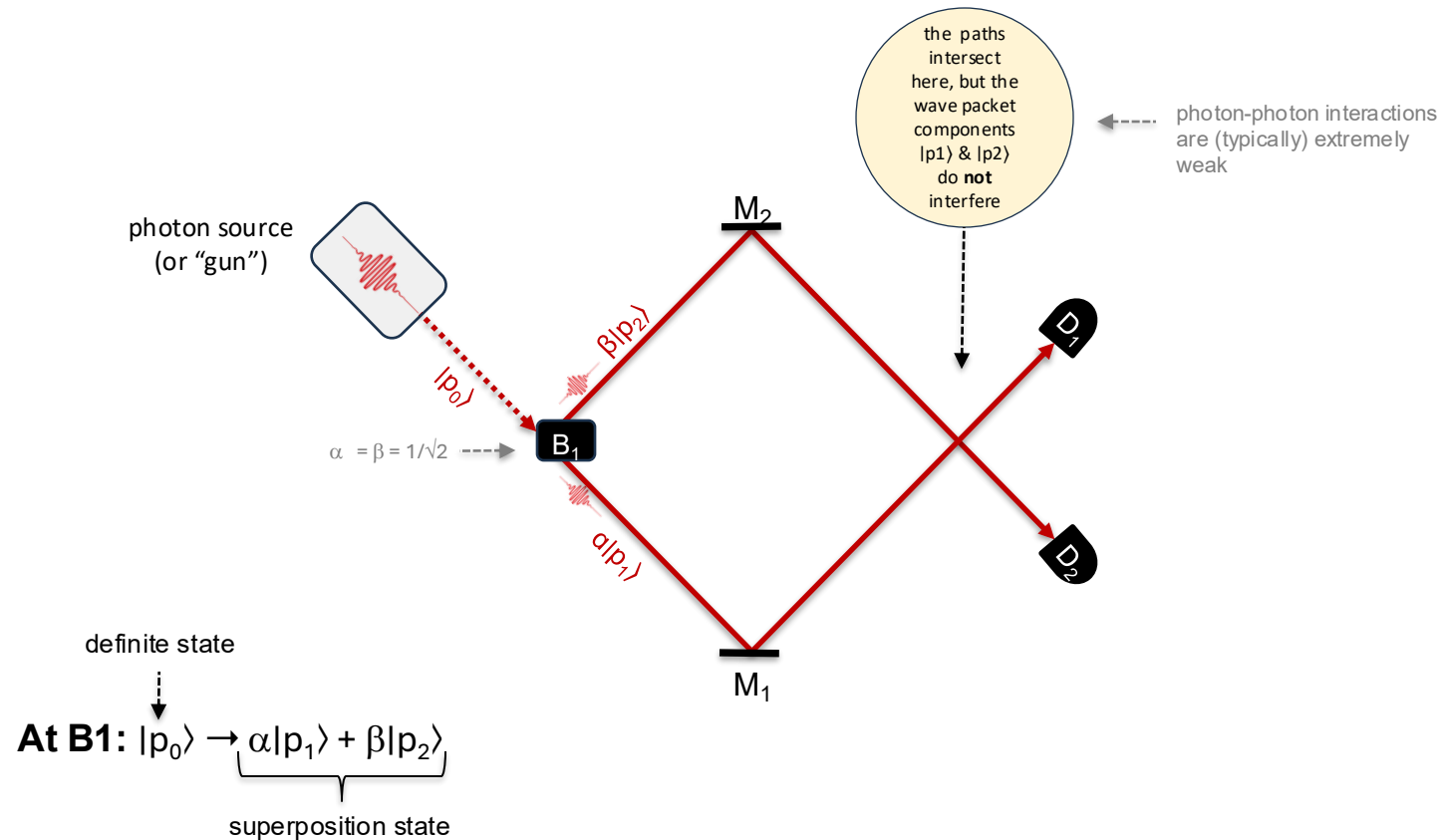
electron orbitals are representations of the wavefunctions of electrons bound to a nucleus

“where” is the electron?



Single photon, 2 paths

“where” is the photon?

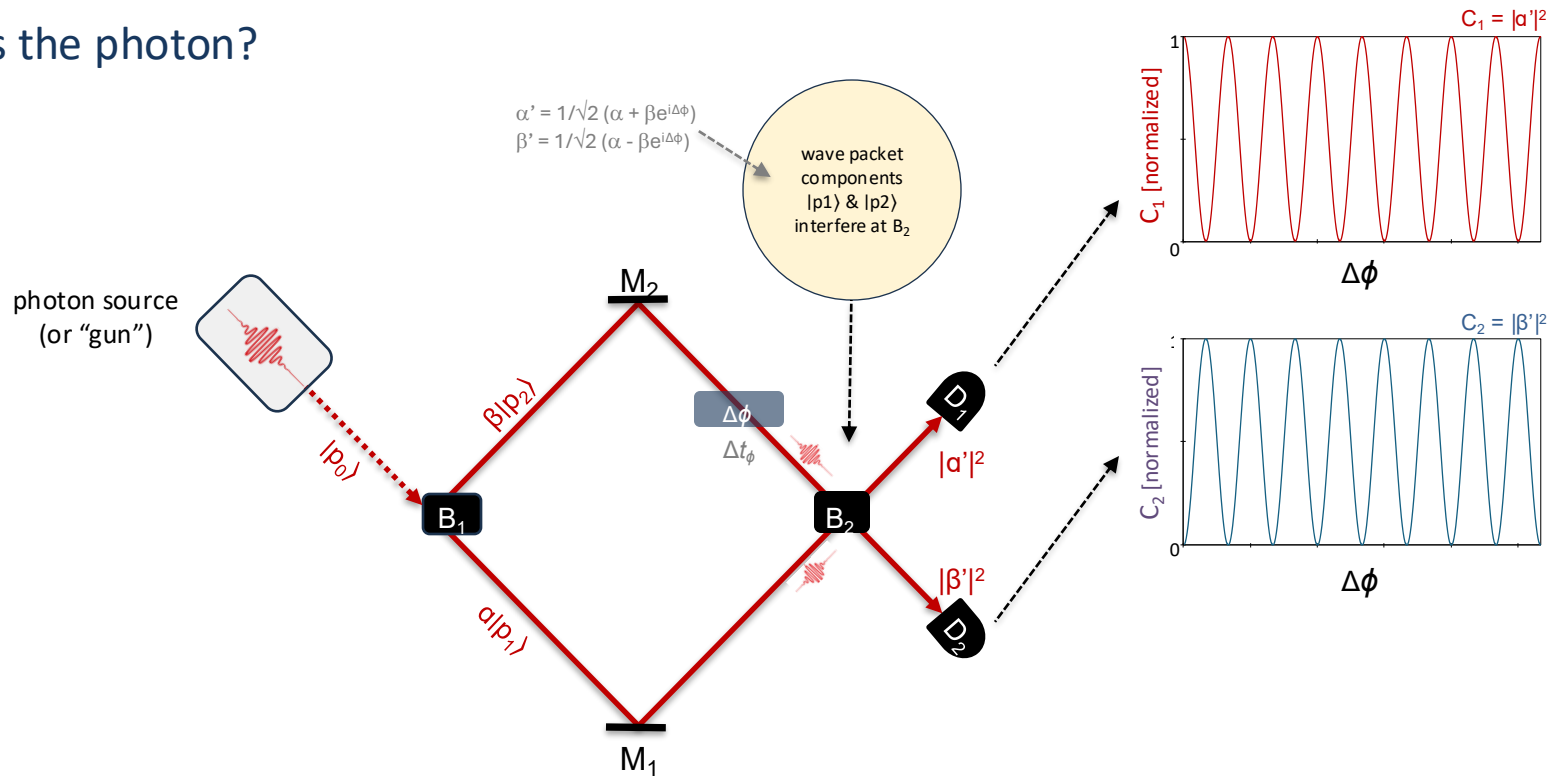


- p_n = photon path n
- α & β = probability densities
- $|\alpha|^2$ & $|\beta|^2$ = probabilities
- $|\alpha|^2 + |\beta|^2 = 1$
- B = 50:50 beam splitter
- M = mirror
- D = single-photon detector

- each photon ‘explores’ both paths 1 & 2 simultaneously
- it remains in this ‘superposition of paths’ until the wave function collapses at D_1 or D_2
- since B_1 is a 50:50 beam splitter, the probability of each photon being detected at D_1 is 50% and also at D_2 is 50%
- paths 1 & 2 are ‘**distinguishable**’ by detection at D_1 or D_2 , respectively

Single photon self-interference

“where” is the photon?



- p_n = photon path n
- α & β = probability densities
- $|\alpha|^2$ & $|\beta|^2$ = probabilities
- $|\alpha|^2 + |\beta|^2 = 1$
- B = 50:50 beam splitter
- M = mirror
- D = single-photon detector

- $\Delta\phi$ = phase shift (either a 'detour' that extends the length of path 2, or a material such as glass that slows down the wave packet component in path 2)
- C_n = photon count rate [Hz] at D_n
- D_n = single-photon detector n

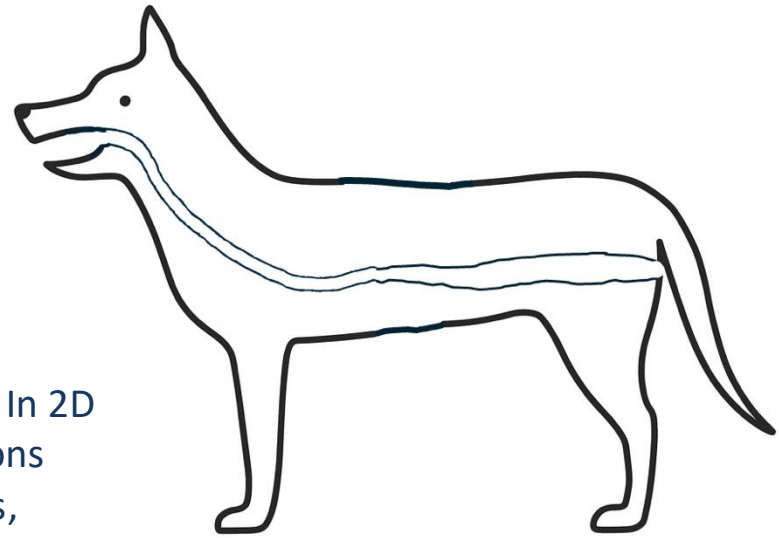
Information accessibility affects physical behavior

e.g. which-path information, photon or electron (in)distinguishability & dimensionality

Why dimensionality is important: In the Portfolio 1 prac assessment you will discuss 2D quantum wells, which are related to 1D quantum wires and 0D quantum dots – all of these are examples of “quantum-confined systems” (i.e. wavefunctions are confined, literally, to 2, 1 or 0 dimensions). The dimensionality constrains what is possible, and leads to unique spatial symmetries, quantization characteristics, densities of states, physical properties, physical behaviors and even the existence or non-existence of physical phenomena – e.g. 1D nanowires can give rise to a unique type of quantized conductivity.

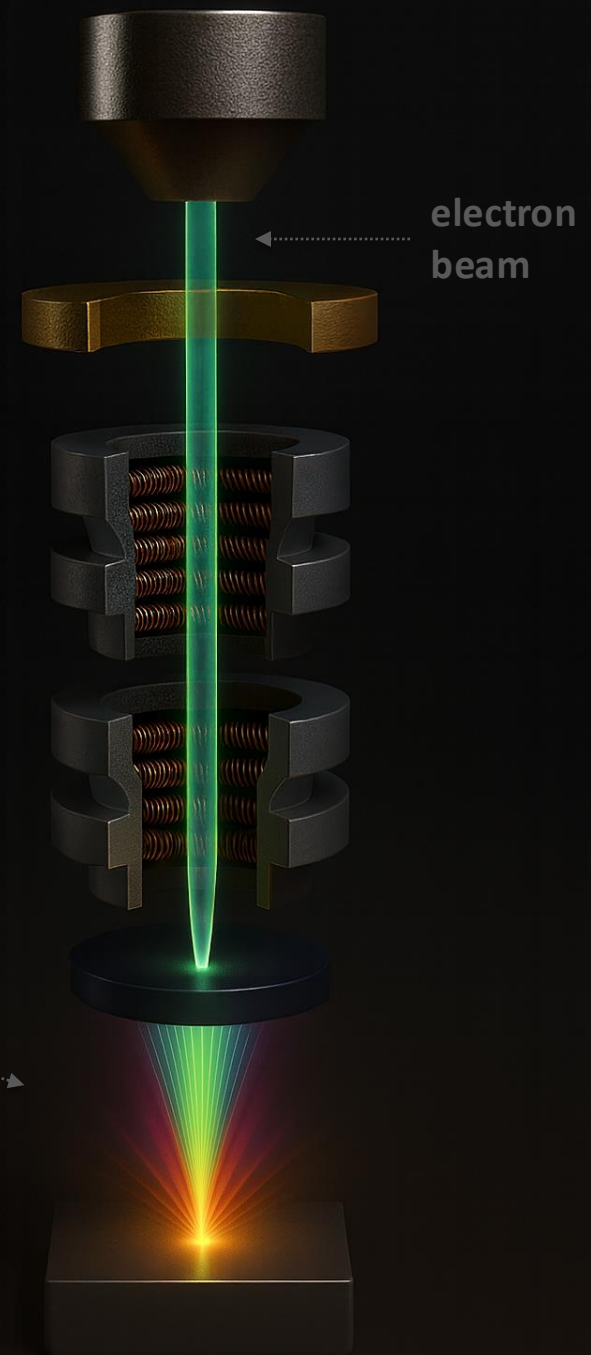
Mental exercise – consider this dimensionality constraint – a biological organism like a dog can't exist in 2 dimensions.

In 3D space the anatomy of a dog is possible because it can accommodate a digestive tract with an entrance and an exit. In 2D space, that arrangement is impossible — certain configurations (and organisms) can't exist. In physics & quantum mechanics, making certain information (in)accessible plays the same role: it changes the “dimensionality” of the space of possibilities the system can explore...

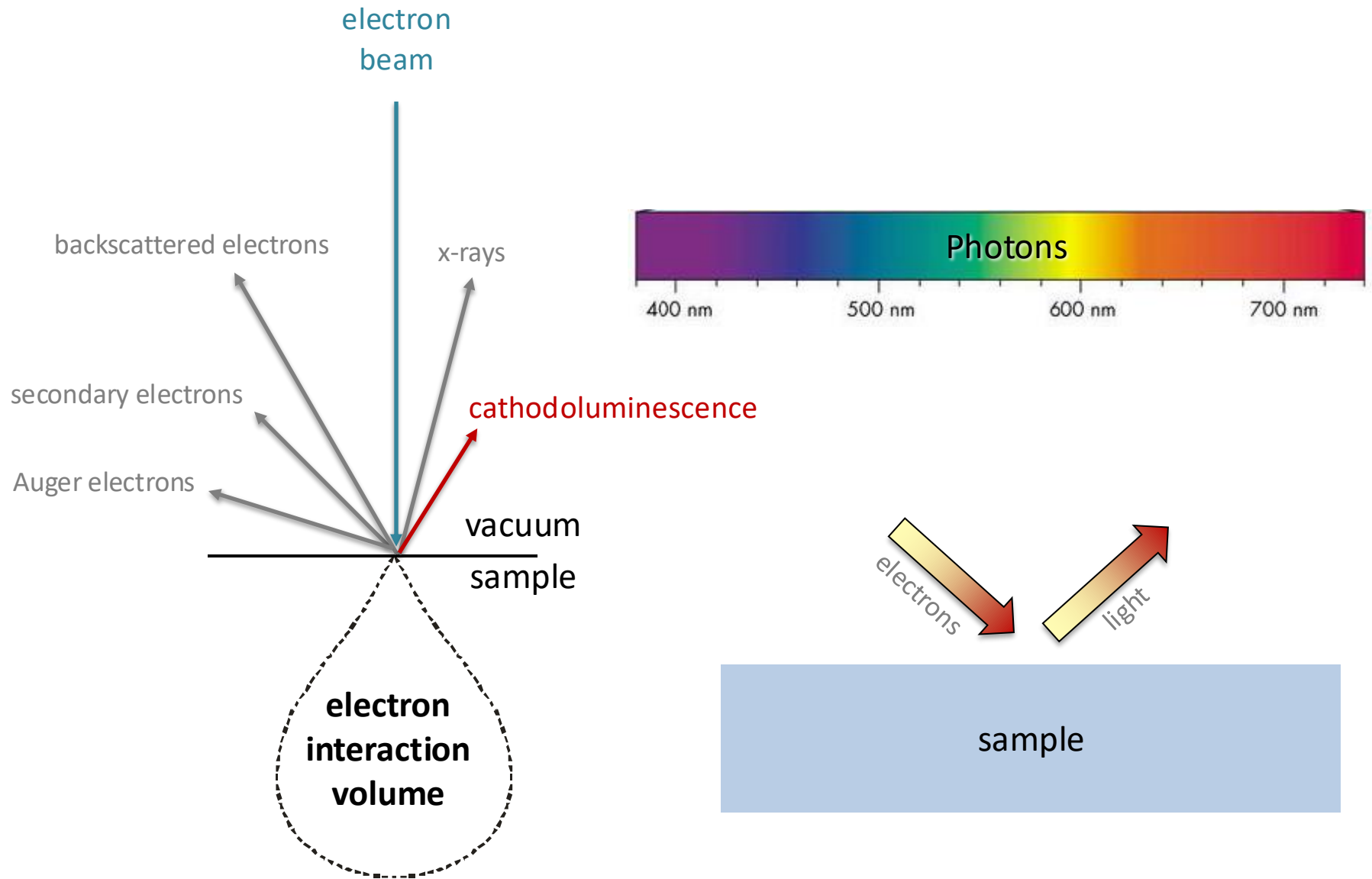


Cathodoluminescence analysis

cathodoluminescence
emitted from the sample

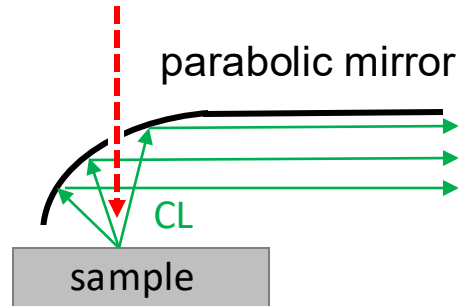


Cathodoluminescence (CL) analysis in a scanning electron microscope



CL: Most common experimental setup

electron beam

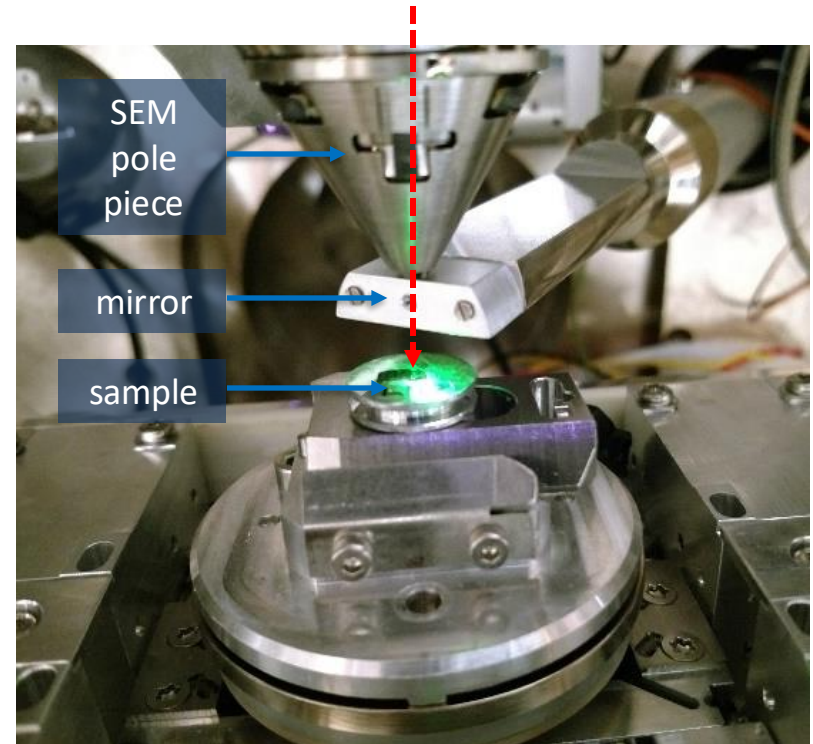
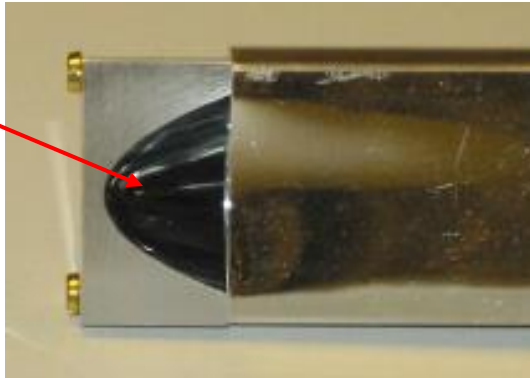


light optics & detectors

electron beam

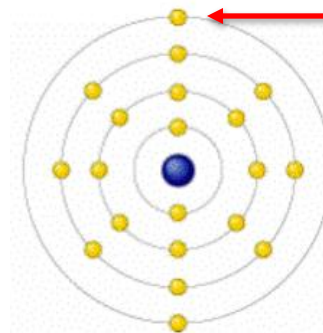
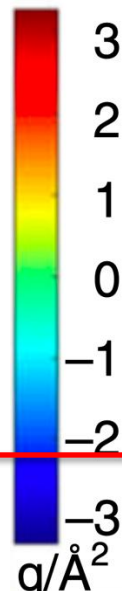
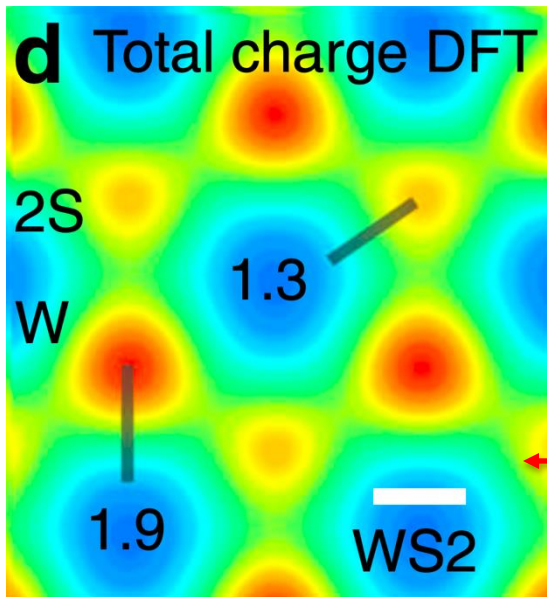
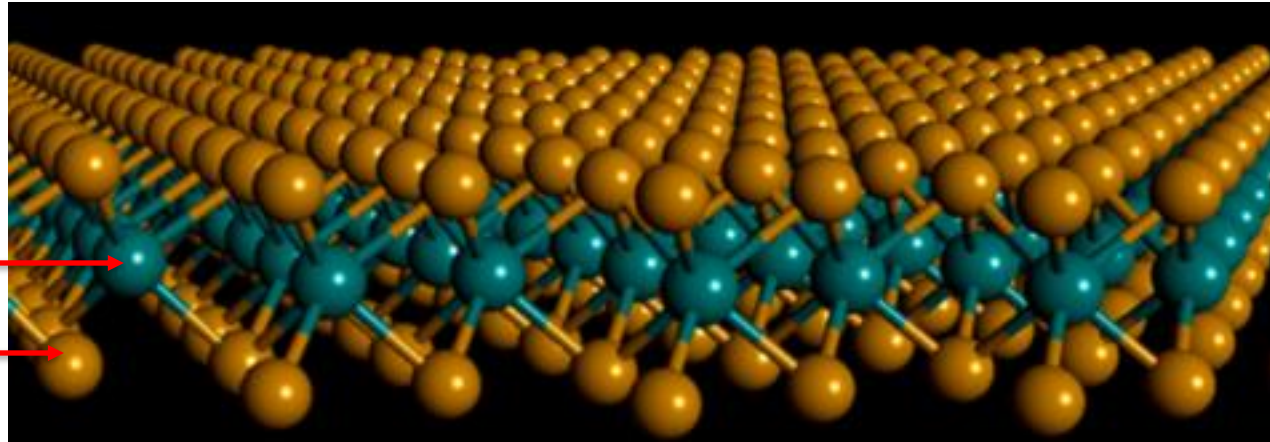
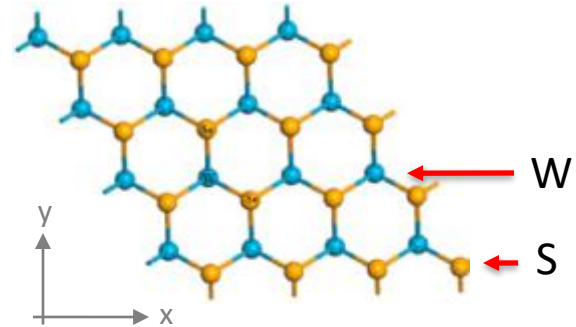
aperture
for electron
beam

parabolic mirror (bottom view)



CL generation: Bond breaking

2D semiconductor WS_2

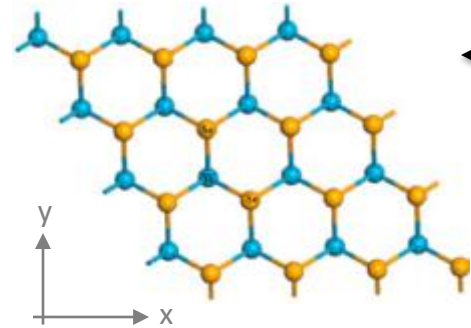


“Outer shell electrons” (a.k.a. “valence electrons”) participate in chemical bonds.

CL involves excitation/ionization of valence electrons.

Valence electrons are “spread out” between the atoms, as illustrated by this “charge density plot” calculated for the 2D material WS_2 .

Charge density & energy bands

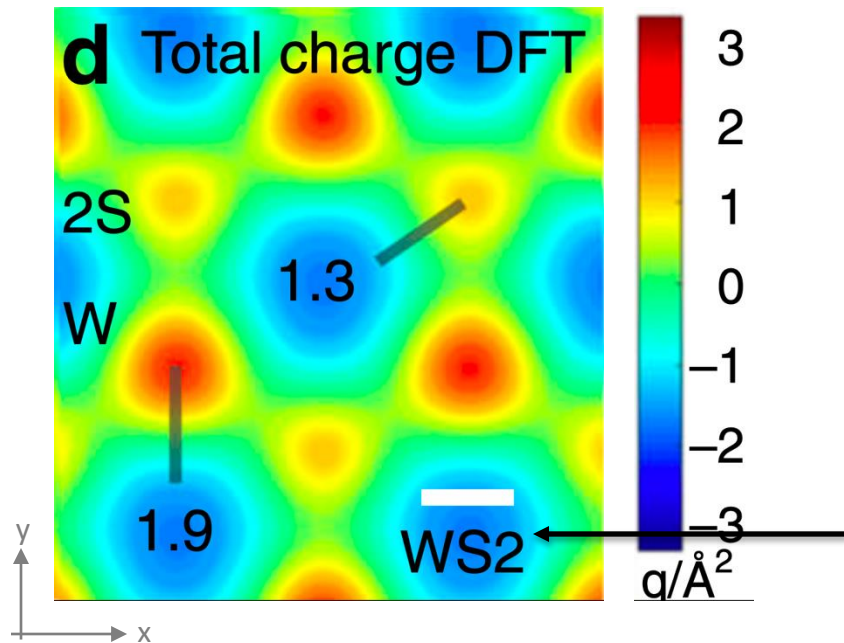


← In a crystal, atoms are arranged in a regular, repeating pattern, creating a **periodic potential** that affects the wavefunction of the electrons.

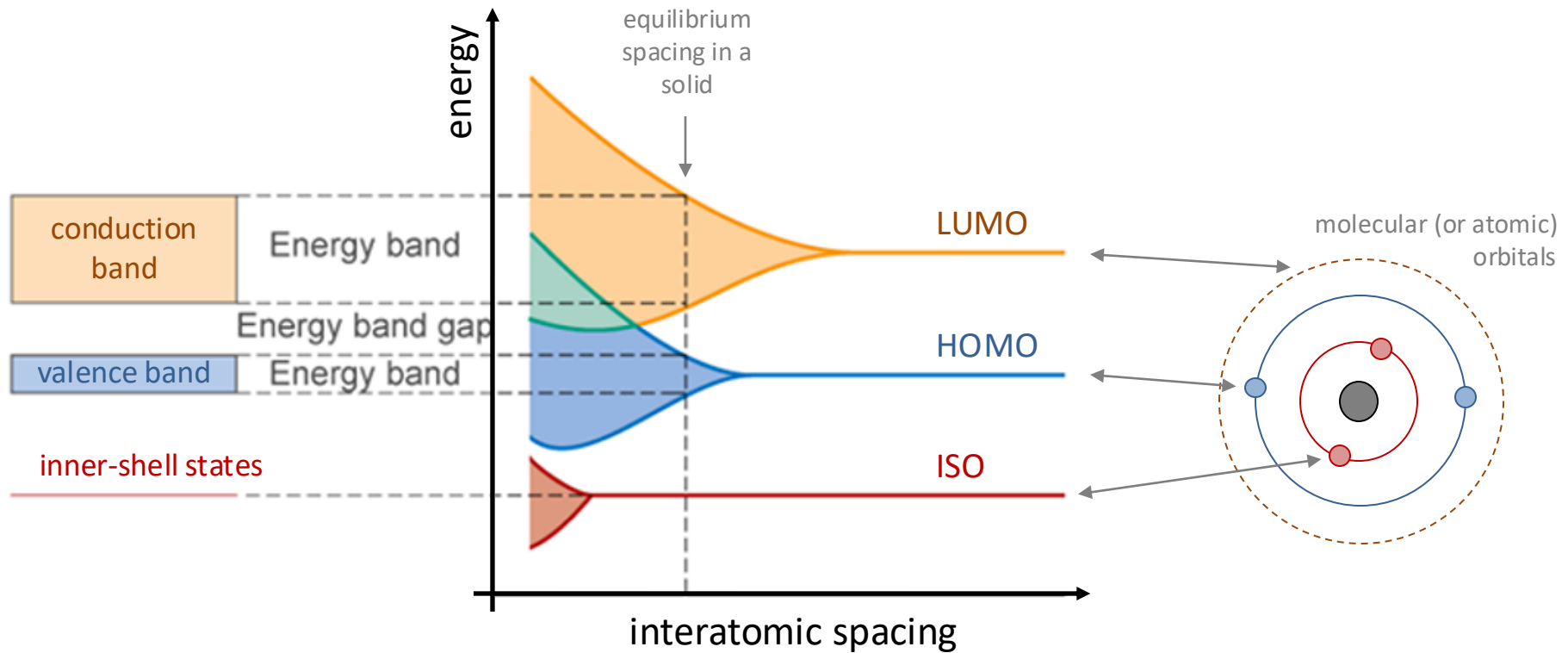
Bloch's theorem: the wavefunction of an electron in a periodic potential can be written as the product of a **plane wave** [NOTE: electrons in the valence band and inner shell states are 'tightly bound to the nuclei' and best thought of as standing waves; a **standing wave** results from the interference of two waves with the same frequency and amplitude traveling in opposite directions] and a function that has the same periodicity as the crystal lattice.

The allowed energy levels of electrons in a crystal form **energy bands**. Each energy band corresponds to a set of Bloch states, where each Bloch state is associated with a specific energy level within the band.

The **charge density** reflects the probability distribution of electrons in the crystal and is directly computed from the Bloch states. Since the crystal potential is periodic, the charge density will also inherit this periodicity.



From atomic orbitals to energy bands



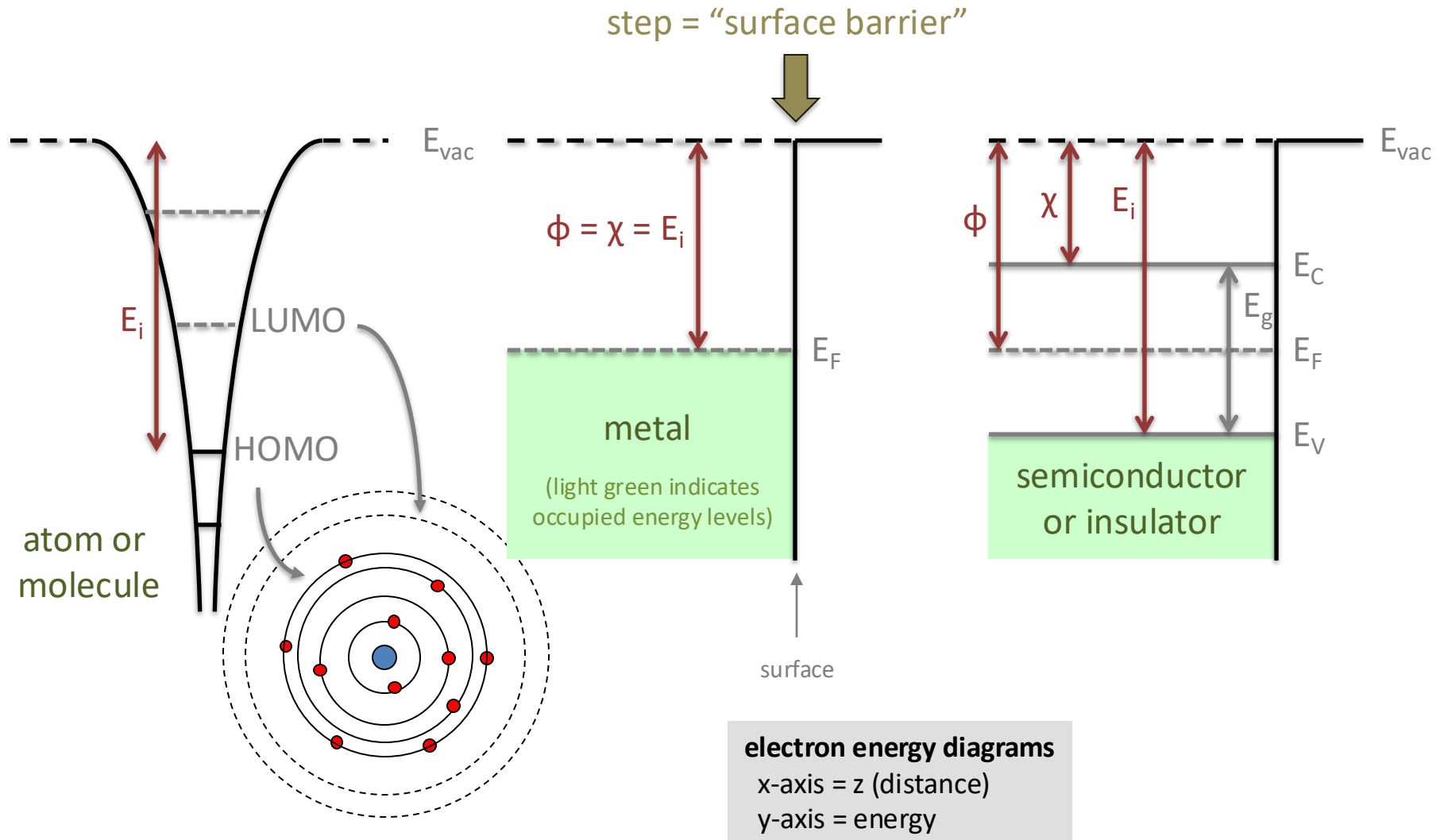
The allowed energy levels in a crystal form **bands** which arise from solutions to the Schrödinger equation with **Bloch states** as the basis.

Pauli exclusion principle plays a role: “no two fermions (such as electrons) can occupy the same quantum state simultaneously”.

LUMO: lowest unoccupied molecular orbital
HOMO: highest occupied molecular orbital
ISO: inner-shell atomic orbital

band gap → dielectric (semiconductors and insulators)
no band gap → metal

Work function & related quantities

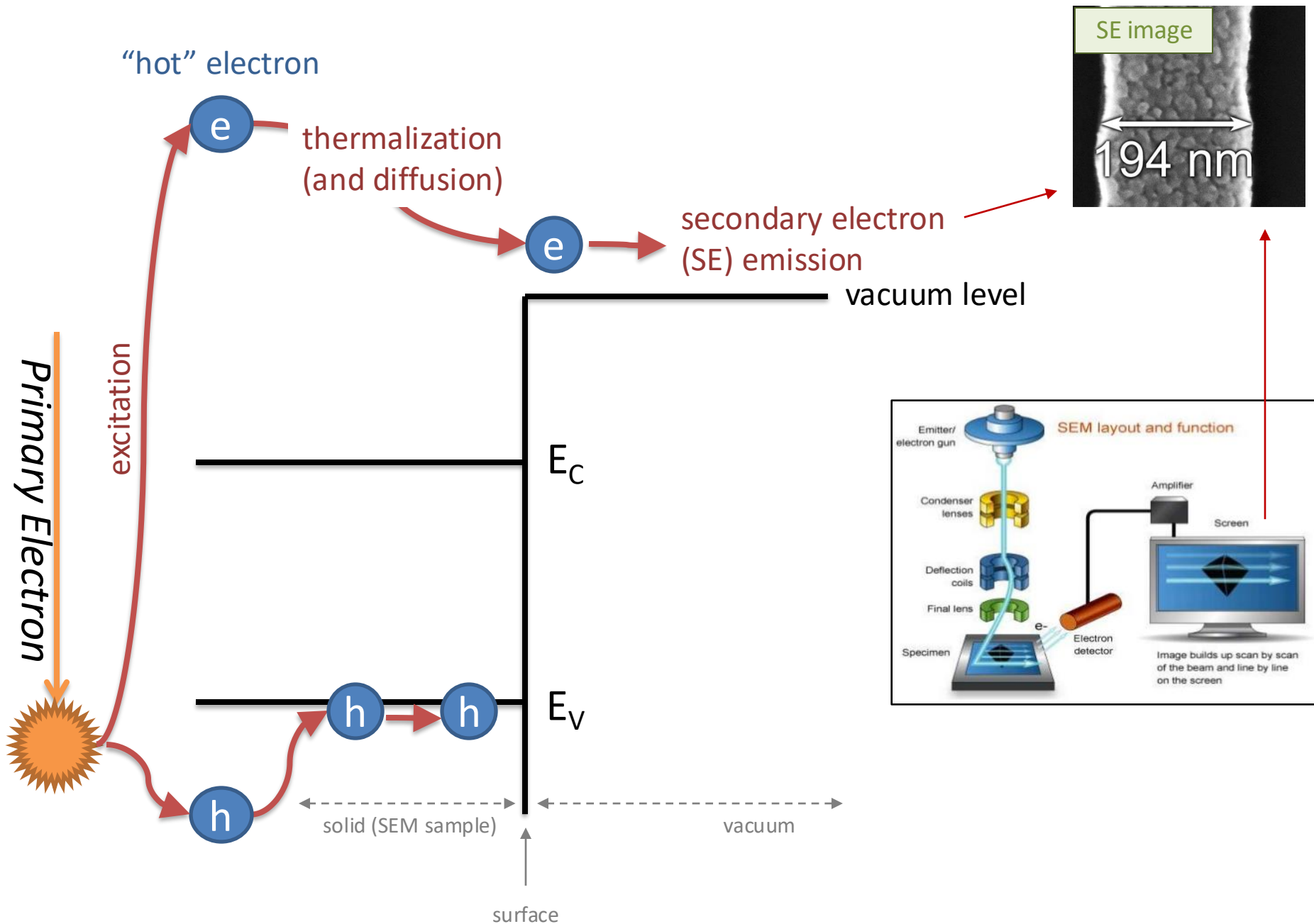


LUMO = lowest unoccupied molecular orbital
HOMO = highest occupied molecular orbital

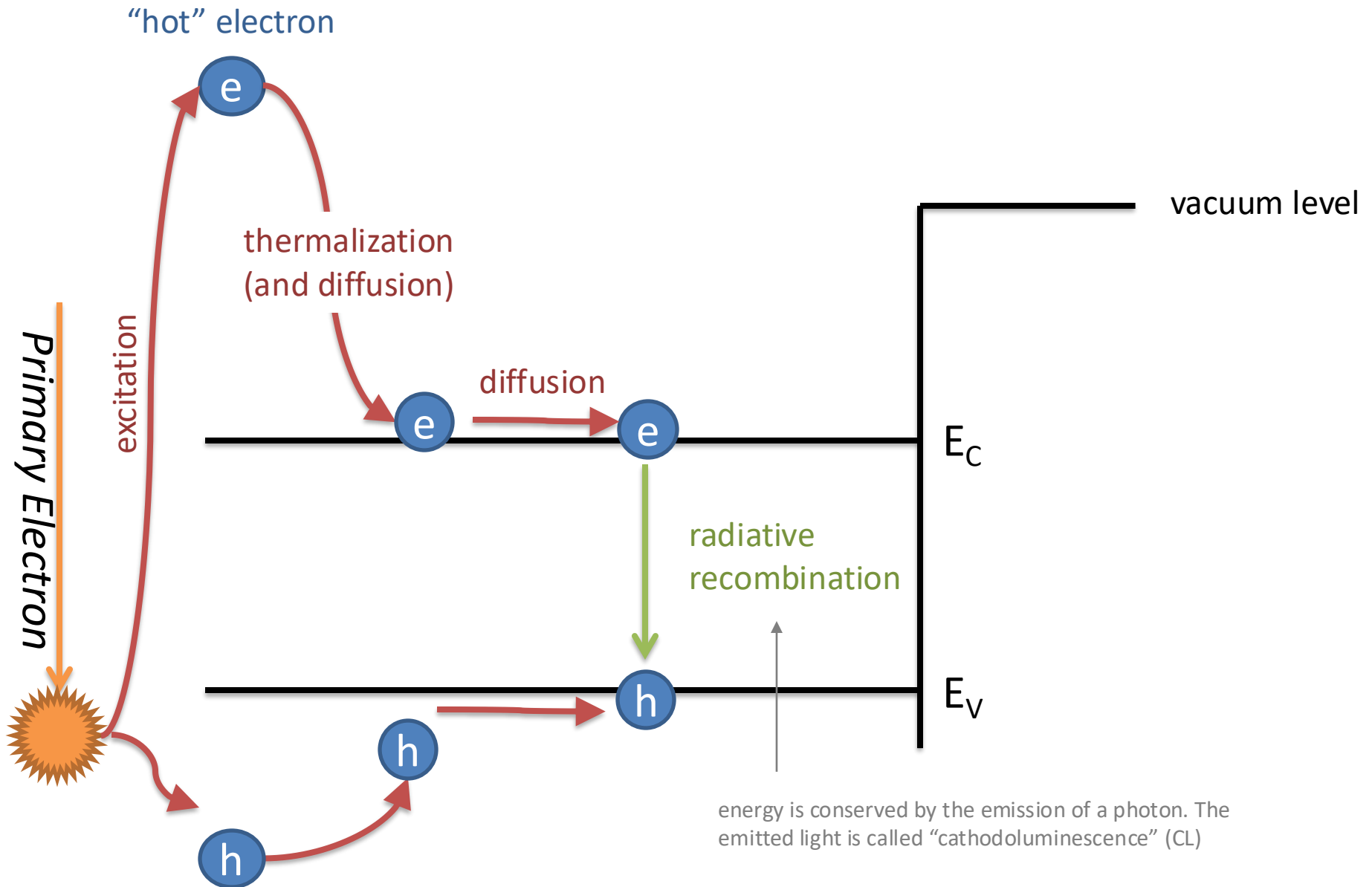
ϕ = work function
 χ = electron affinity
 E_i = ionization energy

E_{vac} = vacuum level
 E_C = bottom of conduction band (CB)
 E_V = top of valence band (VB)
 E_F = Fermi energy

SEM: Electron excitation & emission

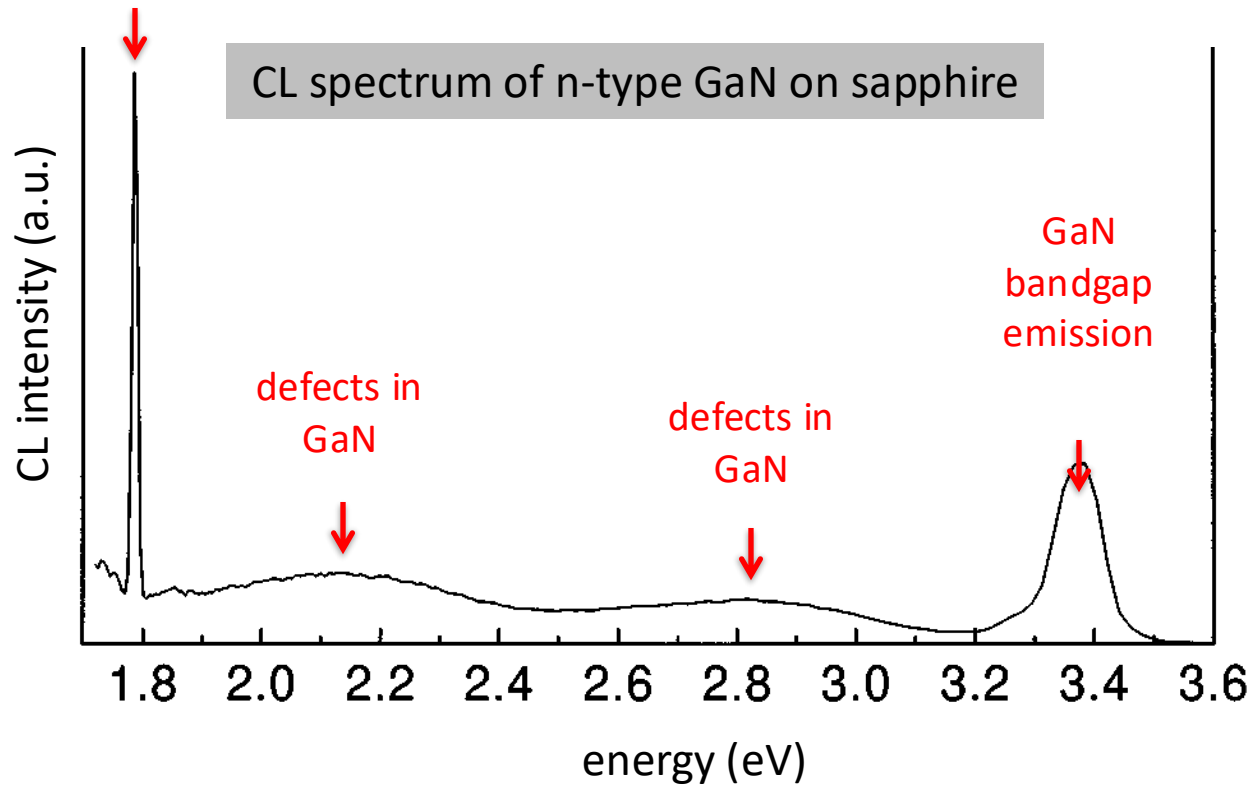


CL generation

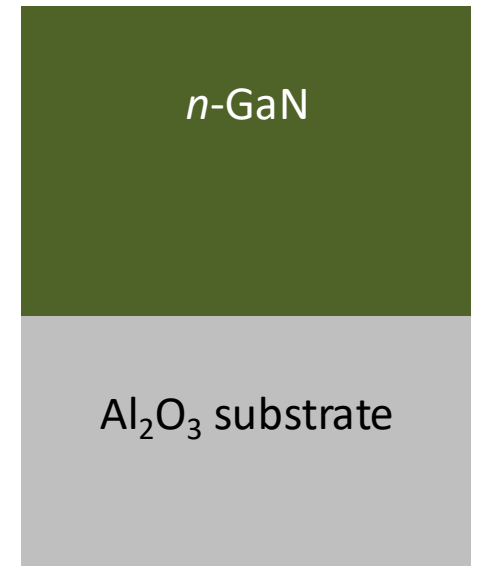


CL spectra

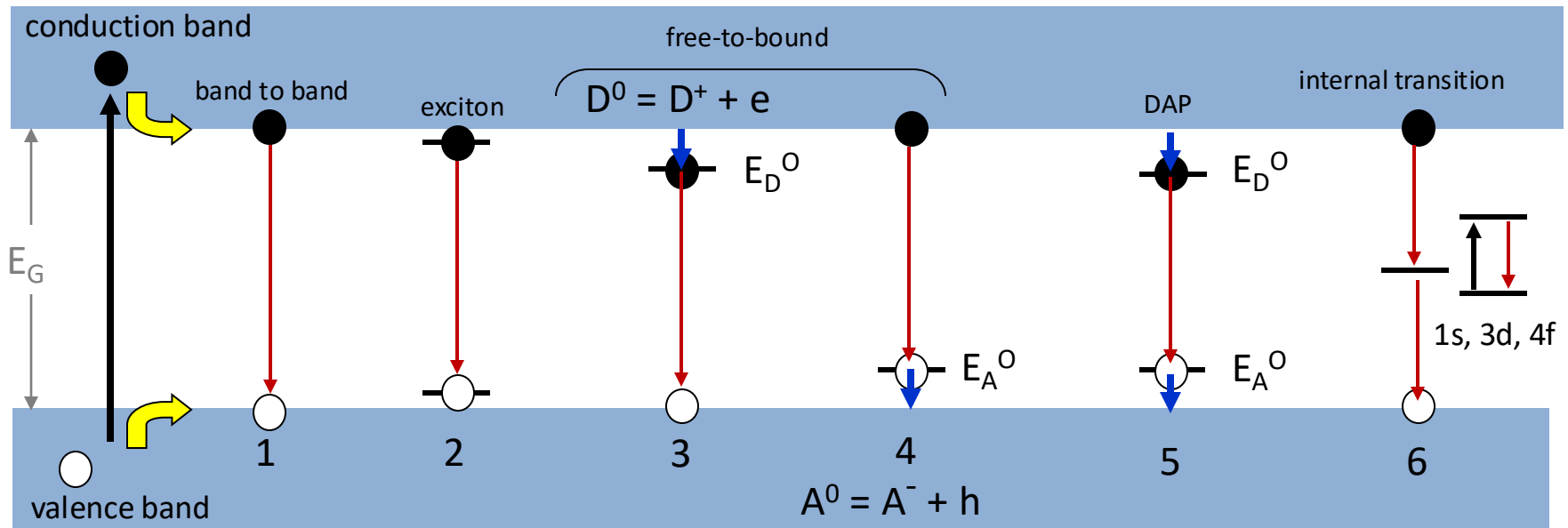
Cr impurities in
sapphire substrate



Sample schematic



Radiative recombination mechanisms

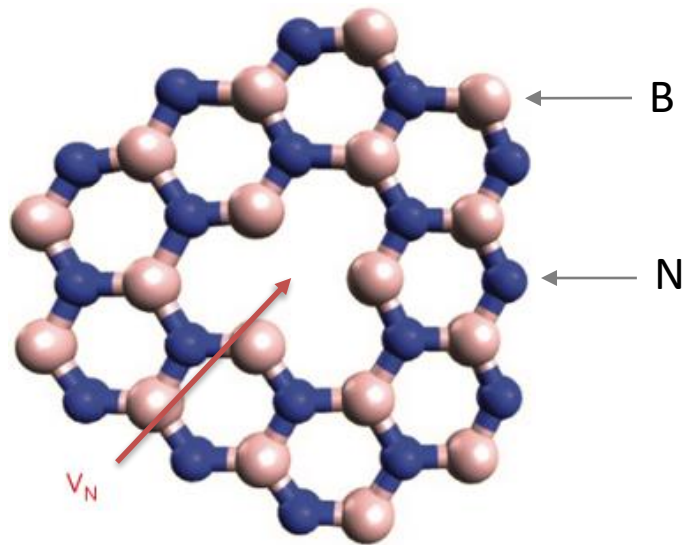


- 1 **band to band** $E_{CL} = E_G$
- 2 **free and bound exciton** $E_{CL} = E_G - E_{EX}$
- 3 **free hole to donor-bound electron** $E_{CL} = E_G - E_D$
- 4 **free electron to acceptor-bound hole** $E_{CL} = E_G - E_A$
- 5 **donor - acceptor pair** $E_{CL} = E_G - E_D - E_A + q^2/(4\pi\epsilon_0\epsilon r)$
- 6 **internal transitions at point defects: e.g., vacancies, interstitial, impurities...**
(see next slide)

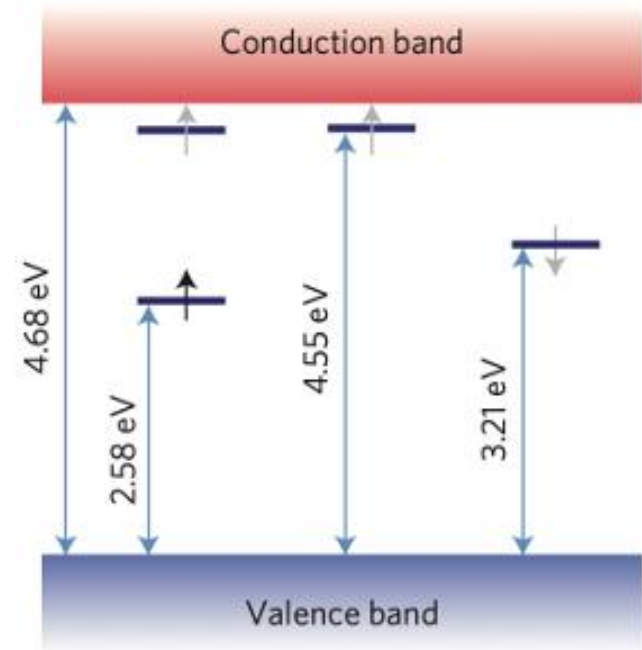
'Point defect' – an example

← From a paper on research done at UTS, in the MAU (i.e., a PhD project)

V_N defect in the hBN lattice



V_N defect states in the band gap of hBN

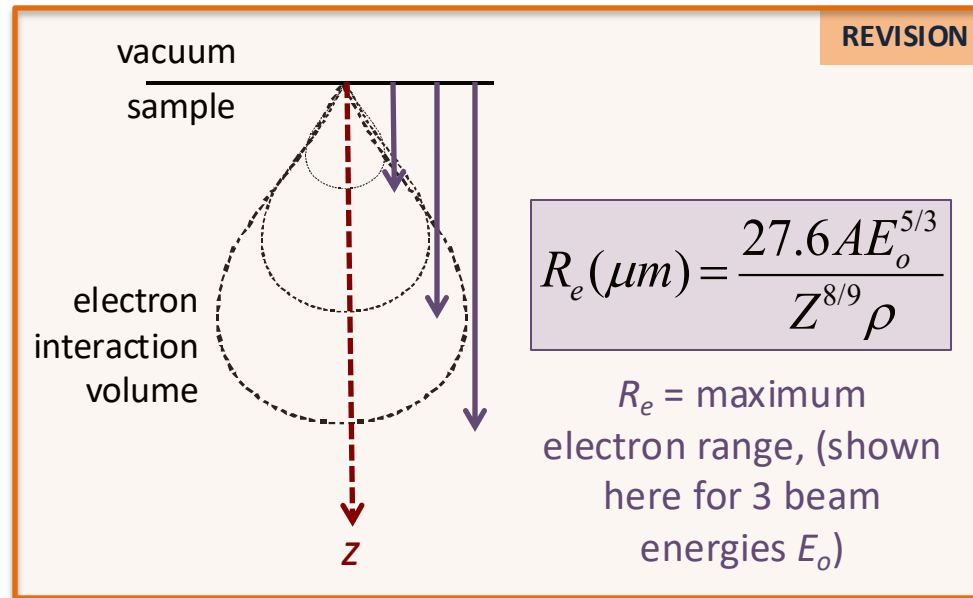


Material: hexagonal boron nitride (hBN) – a 2D material analogous to graphene

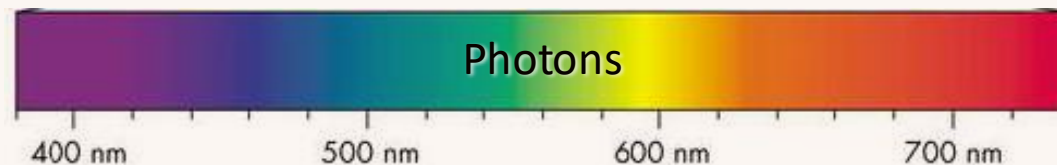
Defect: missing nitrogen atom (nitrogen 'vacancy' V_N)

CL spatial resolution

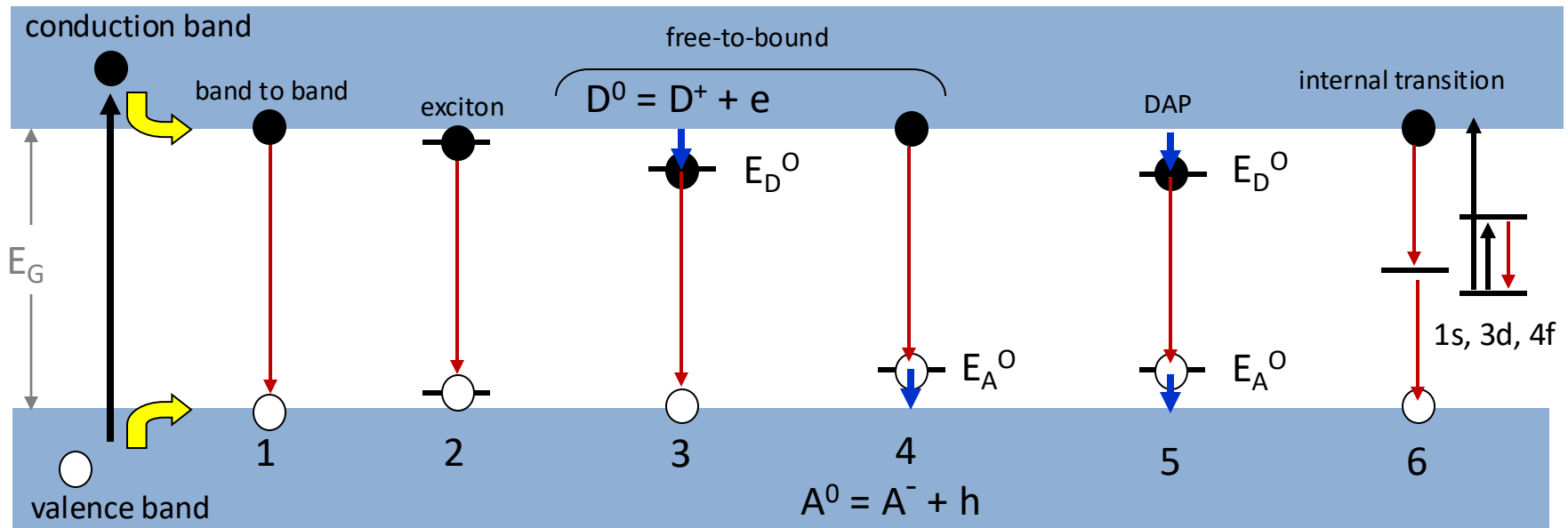
- limited by 3 factors: size of the interaction volume, carrier diffusion and **self-absorption**
- **not** limited by the wavelength of the emitted light



diffraction resolution limit in optical microscopy $\approx \lambda/2$



Absorption of CL by the sample

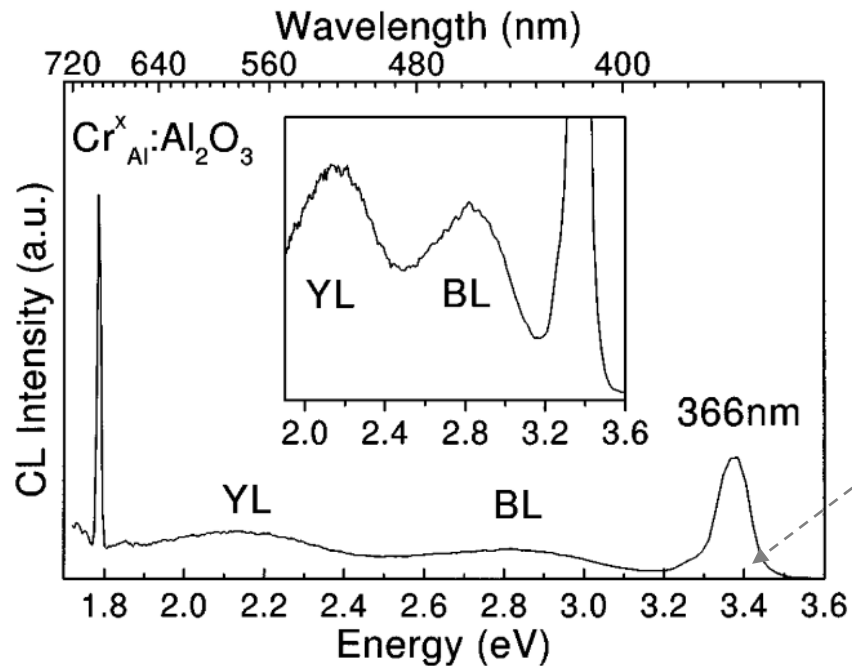


The black arrows represent electron excitation pathways. In CL, the electrons in the sample are excited by the electron beam (i.e., by the so-called “primary electrons”).

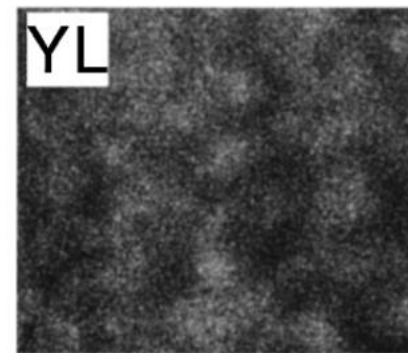
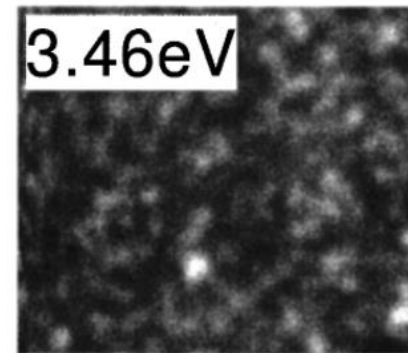
Electrons in the sample can also be excited by photons, including CL photons. This results in absorption of the photons (i.e., some of the CL photons are absorbed by the sample and therefore do not reach a detector).

- 1 **band to band** ← Very efficient CL absorption pathway, because there is a high density of energy states in the conduction band and in the valence bands. However, the photon energy must be $\geq E_G$.
- 6 **internal transitions at point defects** ← The efficiencies of these pathways depend on the defect concentrations (“densities”), and on the photon energy. In high quality semiconductors (i.e., “optoelectronic grade”), these pathways are extremely inefficient, and the resulting absorption is negligible

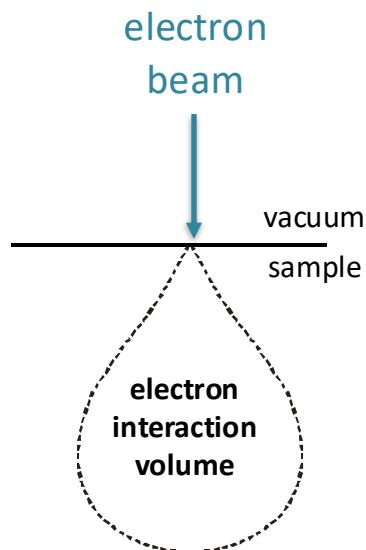
CL spatial resolution: Role of absorption



3.46 eV bandgap emission is absorbed by the sample, hence image resolution is high



All CL emissions are generated throughout the interaction volume. However, the 3.46 eV bandgap emission is absorbed by the sample, and only CL generated near the surface is emitted

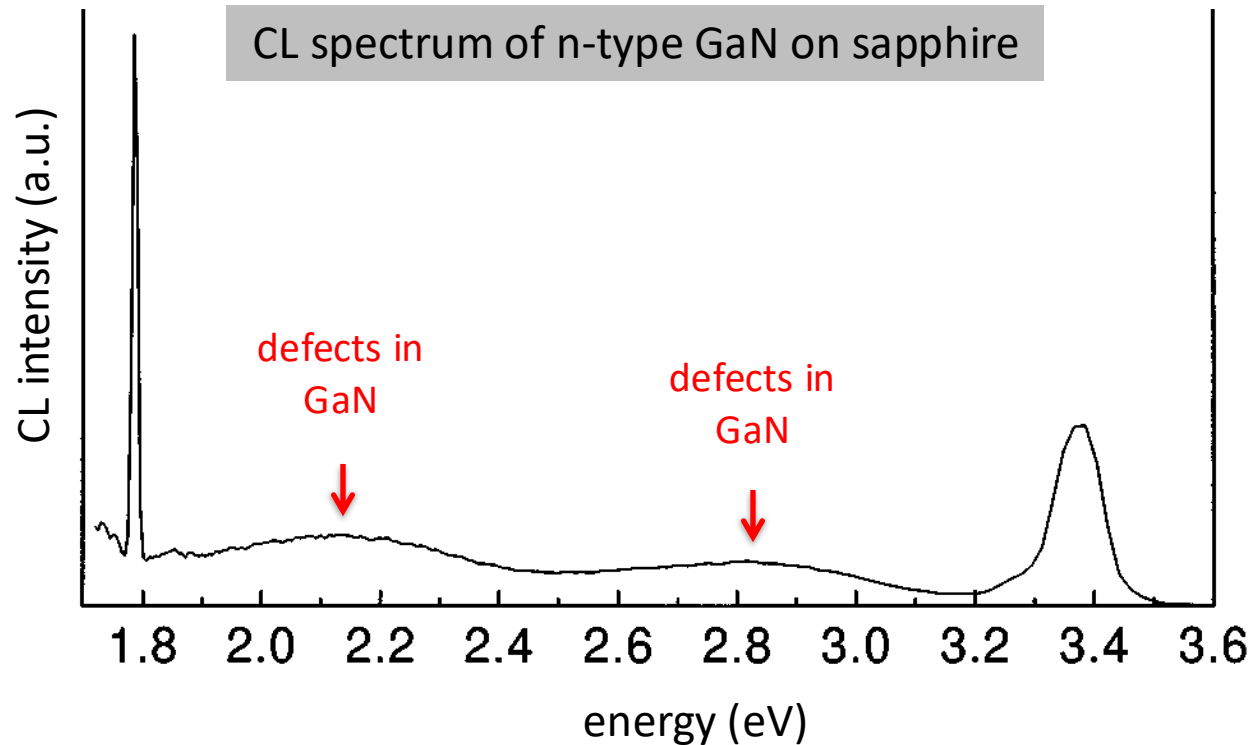


YL defect emission is not absorbed by the sample, hence image resolution is lower (analogous to backscattered electron image resolution)

Electron emission:
**Secondary Electron (SE) imaging in Scanning
Electron Microscopy (SEM)**

Sample class test questions

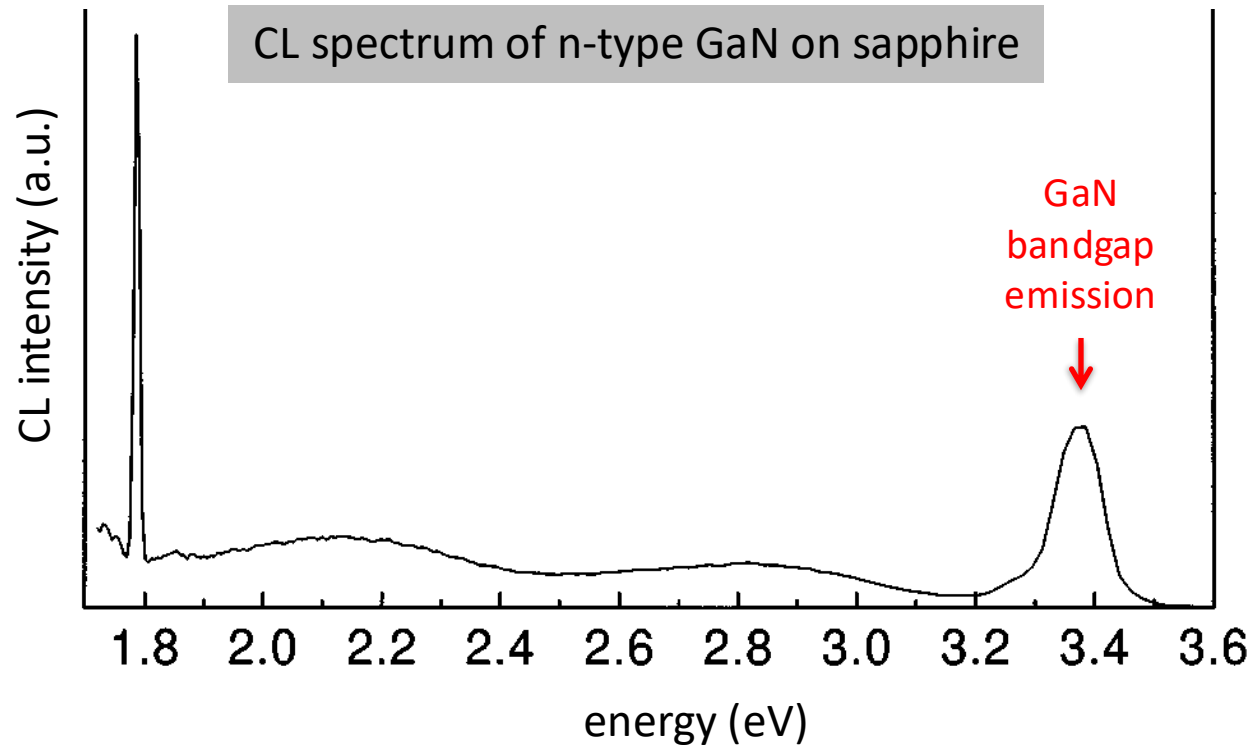
CL 1



Consider this cathodoluminescence (CL) spectrum from the semiconductor GaN. The GaN thickness is 2 microns and the electron beam energy is 30 keV. Which of the following limits the spatial resolution of CL images generated using the CL emissions indicated in red on the spectrum?

- a) Electron beam energy.
- b) Energy of the CL photons.
- c) Absorption of light by GaN.
- d) Wavelength of the CL photons.
- e) Electron-hole pair recombination rate.
- f) Atomic weight of the defects responsible for the CL emissions.
- g) Bandgap of GaN.

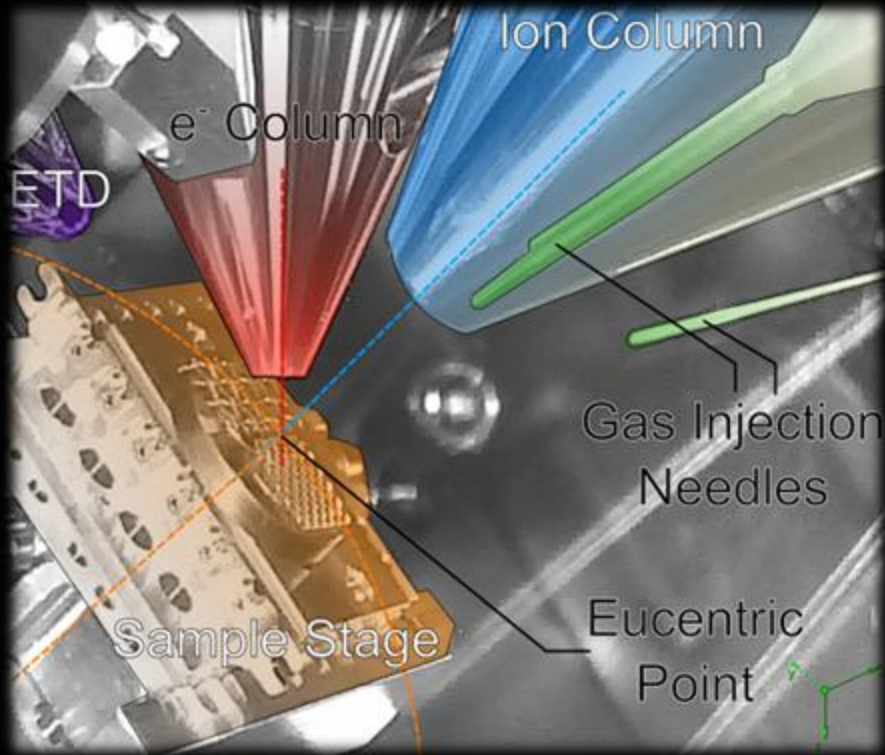
CL 2



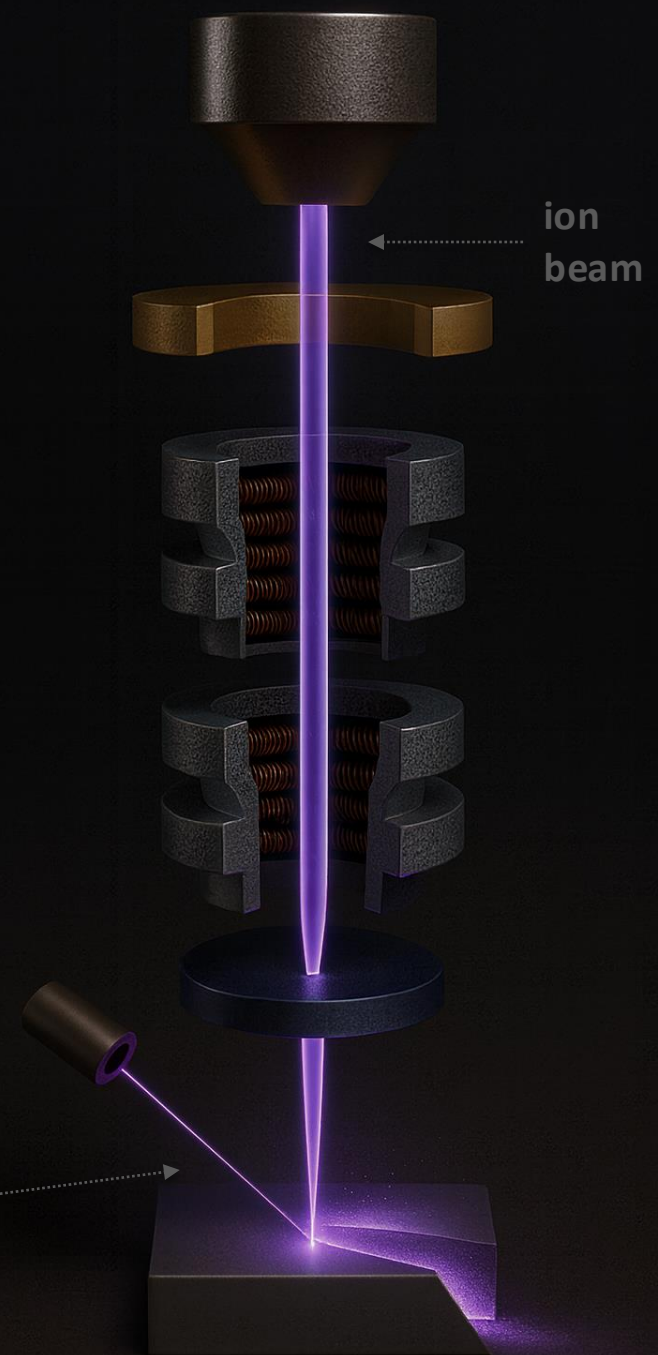
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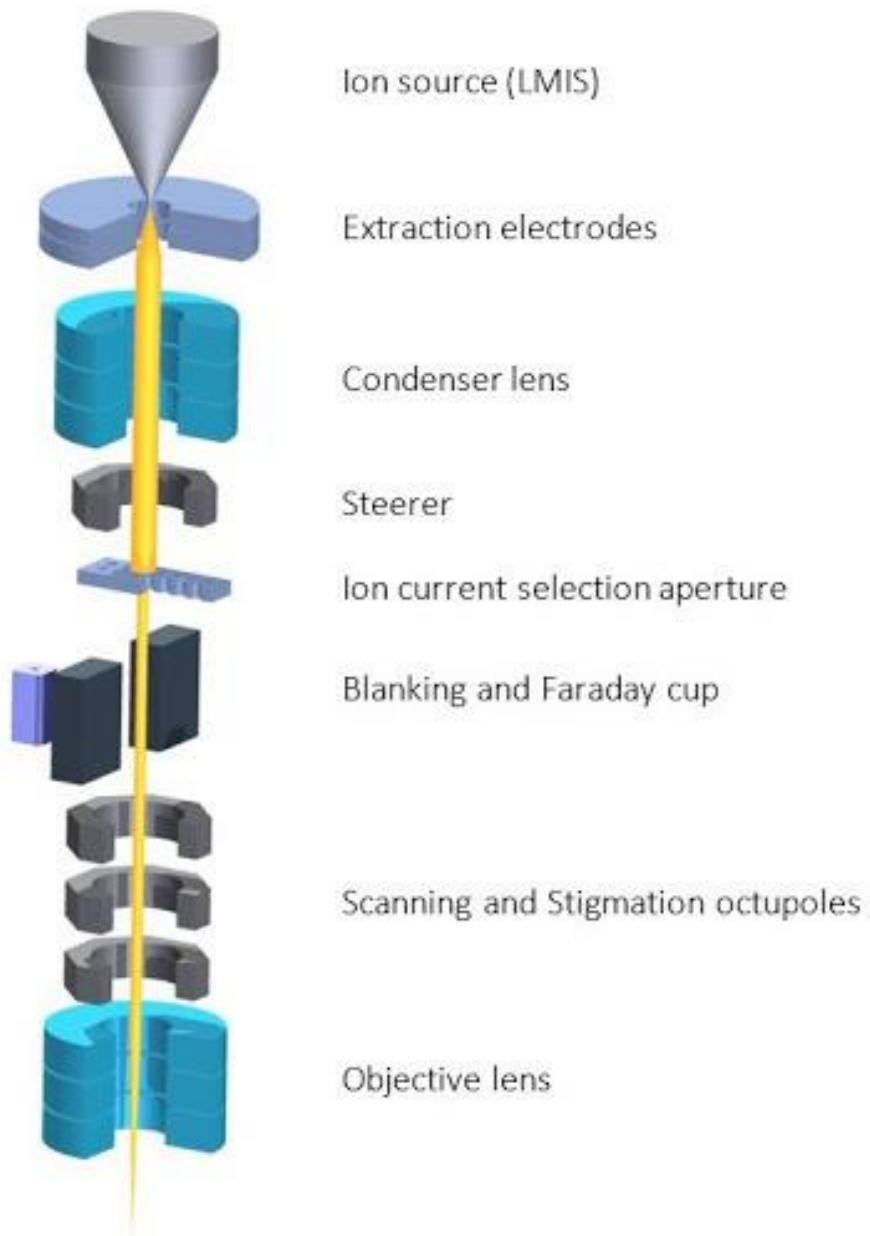
Focused ion beam milling



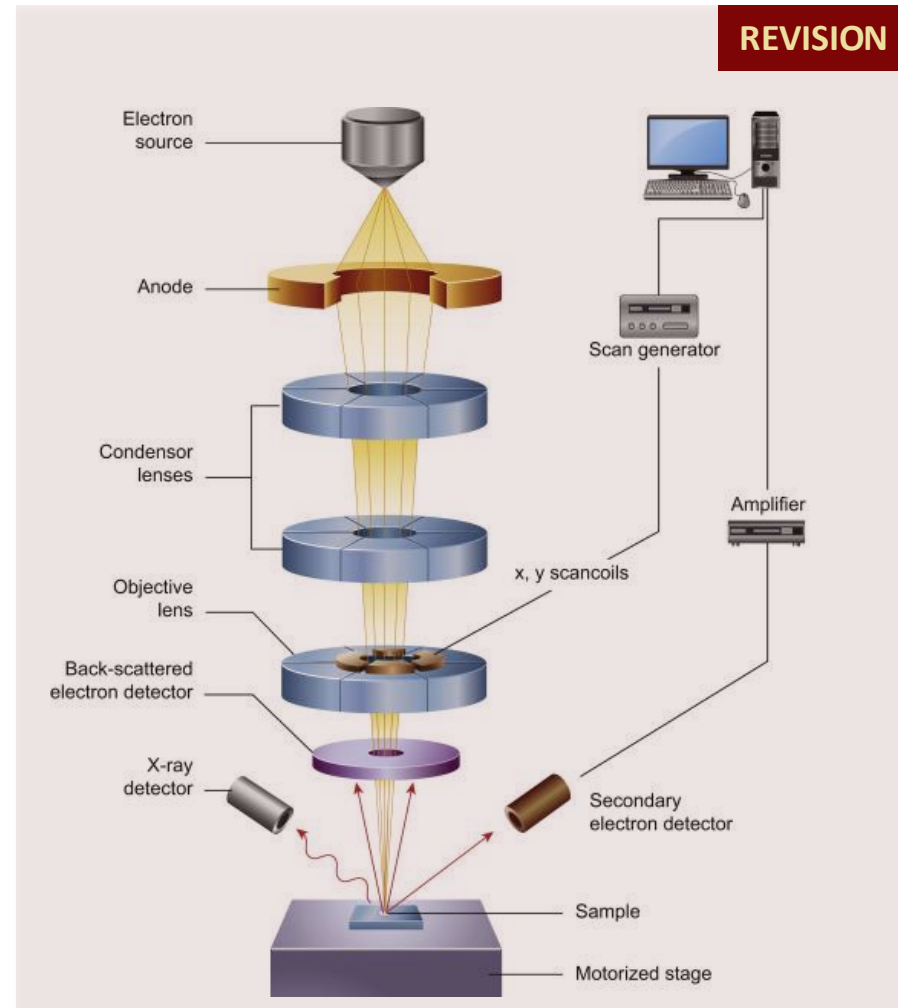
secondary electrons or ions generated by the ion beam can be used for imaging



Focused ion beam (FIB) column



Similar to SEM column

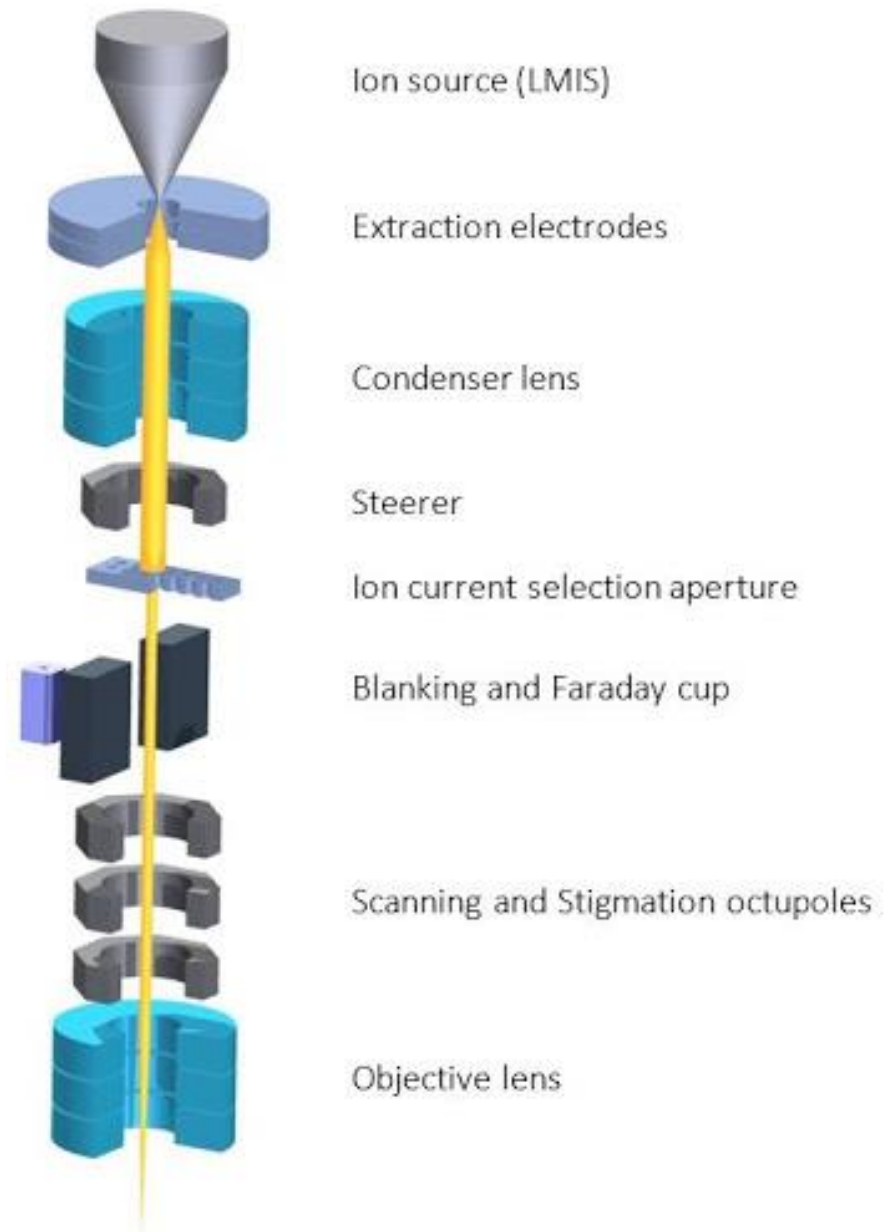


Focused ion beam (FIB) column

Magnetic lenses are used in SEM.

Electrostatic lenses are used in FIB instruments.

... lets briefly review some basics of electromagnetism ...



Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$
$$= \mathbf{F}_E + \mathbf{F}_B$$

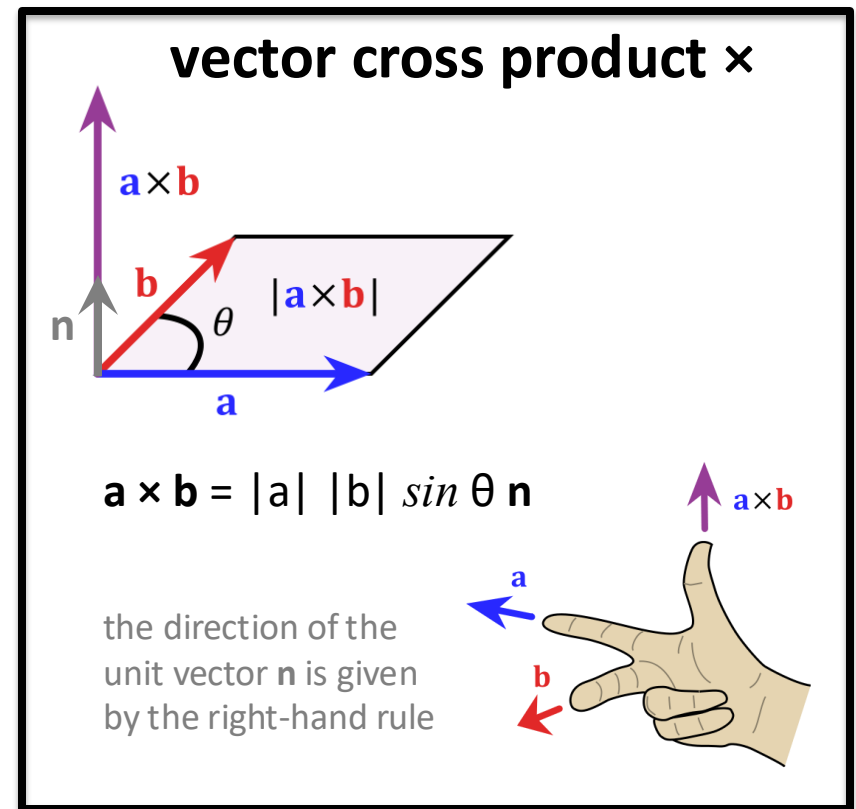
- $\mathbf{E}$ = electric field [units: $\text{NC}^{-1} = \text{Vm}^{-1}$]
- $\mathbf{B}$ = magnetic field [$\text{T} = \text{NC}^{-1}\text{m}^{-1}\text{s}$]

- $q\mathbf{E} = \mathbf{F}_E$ = electric force [N]
- $q\mathbf{v} \times \mathbf{B} = \mathbf{F}_B$ = magnetic force [N]
- $\mathbf{F}$ = “Lorentz force” [N]

- if $q = 0$, then $\mathbf{F}_E = \mathbf{F}_B = 0$ (no force unless particle is charged)
- if $\mathbf{v} = 0$, then $\mathbf{F}_B = 0$ (no magnetic force unless particle is moving)

- Electric fields are created by charged particles & exert forces on charged particles.
- Magnetic fields are created by & exert forces on moving charged particles.
- Unlike an electric field, a constant magnetic field changes the direction of motion of a charged particle, but does not change the speed of the particle.

Typically, magnetic fields are used to focus electron beams and electric fields to focus ion beams



Focused ion beam (FIB) column

Ion source: LMIS or GFIS or plasma

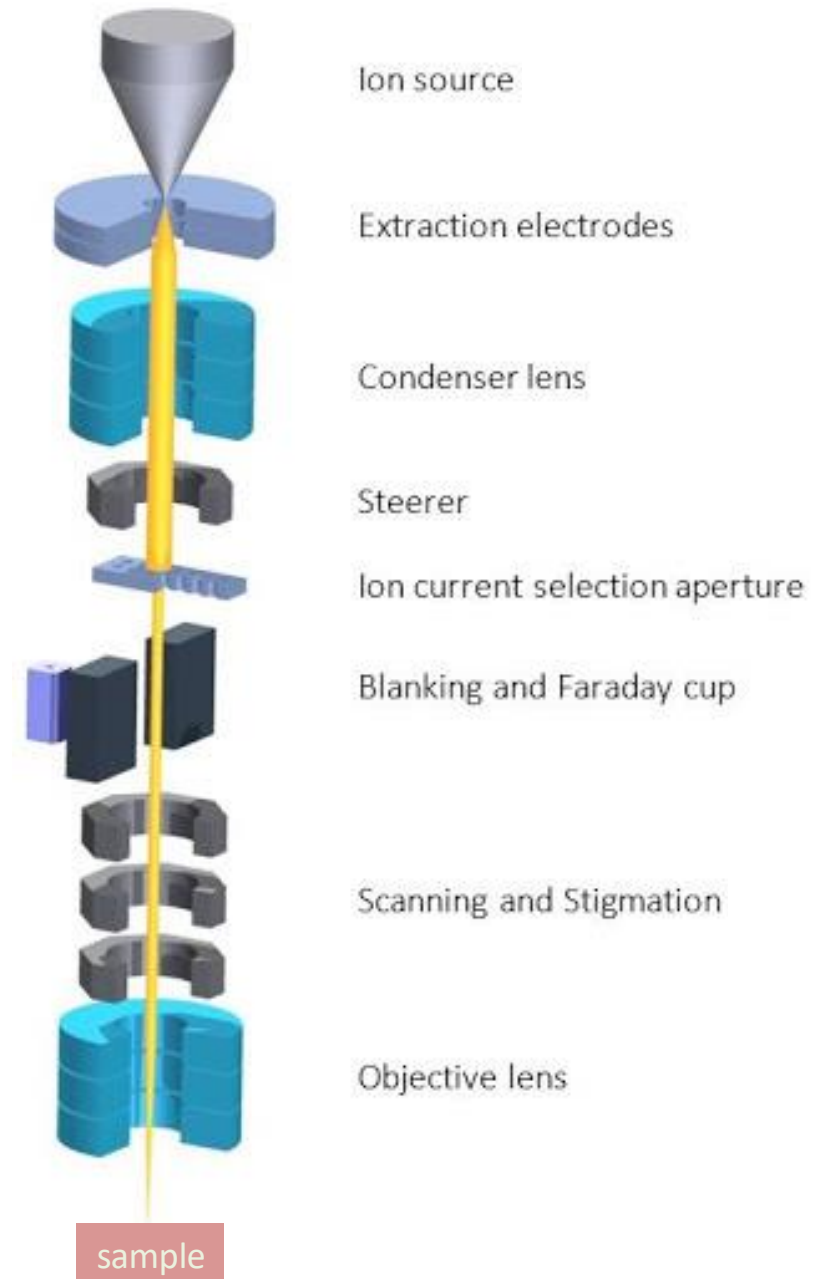
Condenser lens (and aperture) used to shape beam & reduce current

Wein filter (and slit or aperture) sometimes used to select the ion species [not shown here]

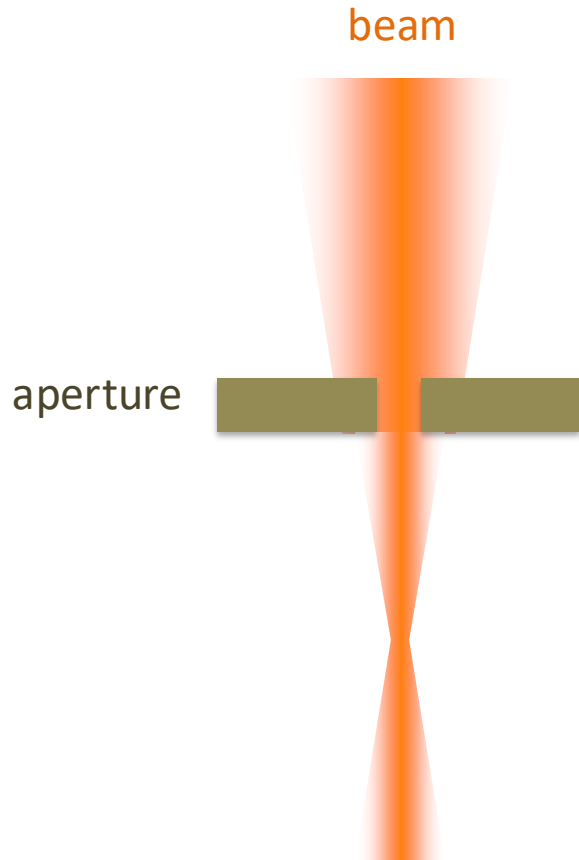
Beam blanker & Faraday cup

Electrodes used to correct astigmatism (beam shape) & to scan the ion beam

Objective lens used to focus the beam onto the sample



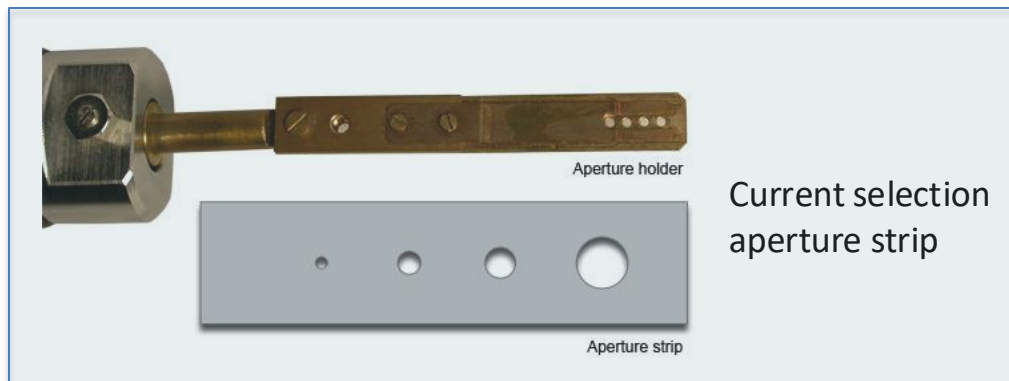
Condenser lens & aperture



Focal length of condenser lens (located above the aperture) determines the fraction of the beam that is transmitted through the aperture...

... the transmitted beam then travels to the objective lens.

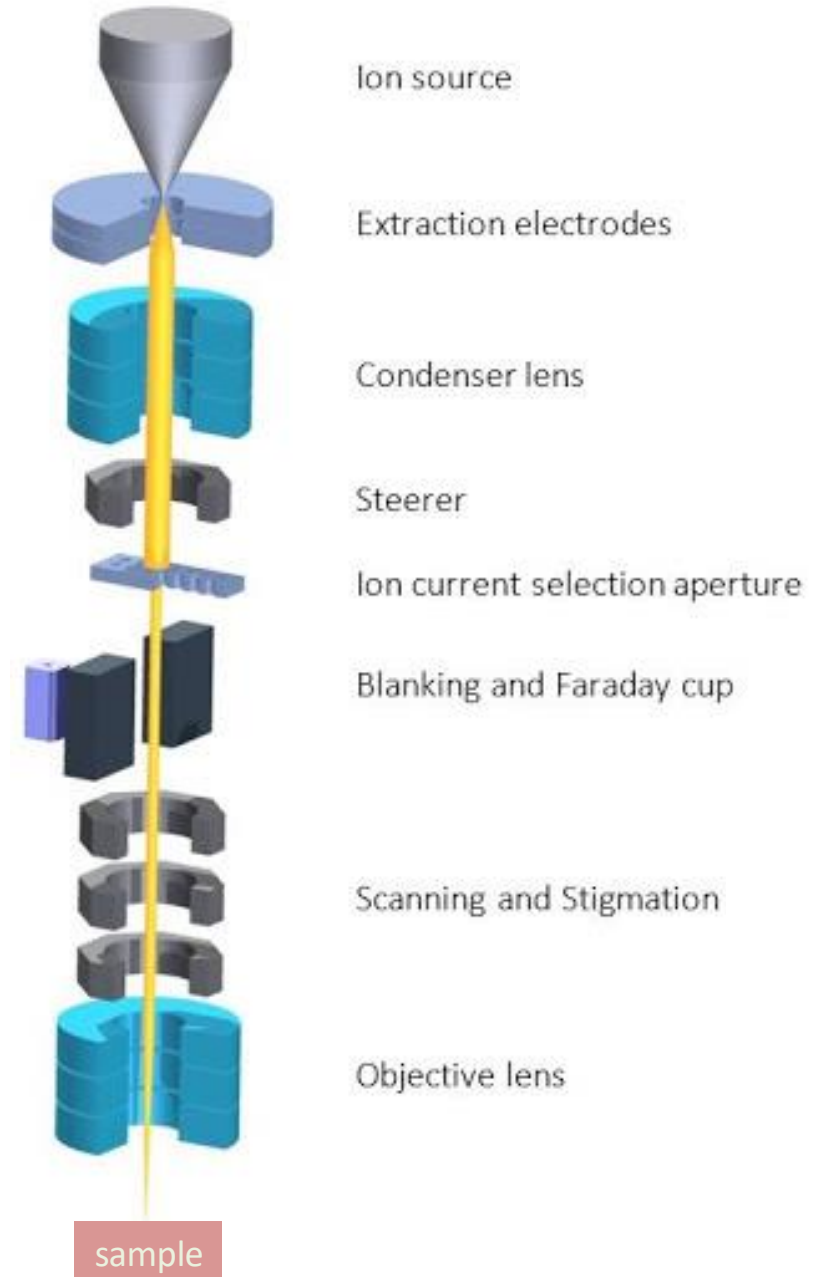
The ions rejected (blocked) by the aperture are high angle ions which exacerbate spherical aberrations – hence, a reduction in beam current made by the objective lens (and aperture) reduces the beam diameter at the sample.



These principles also apply to SEM columns: A condenser lens and an aperture is used to control the electron beam current, which affects resolution.

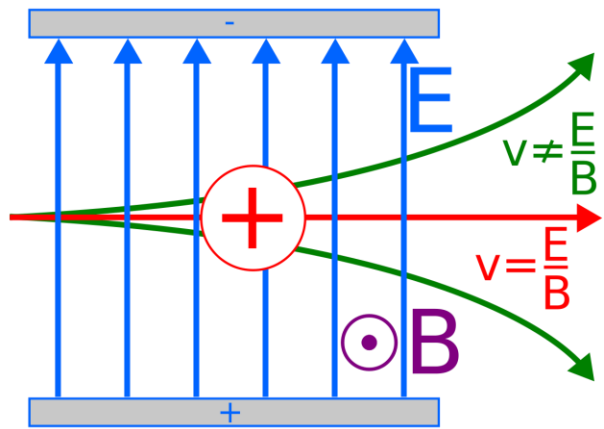
Focused ion beam (FIB) column

Wein filter (and slit or aperture)
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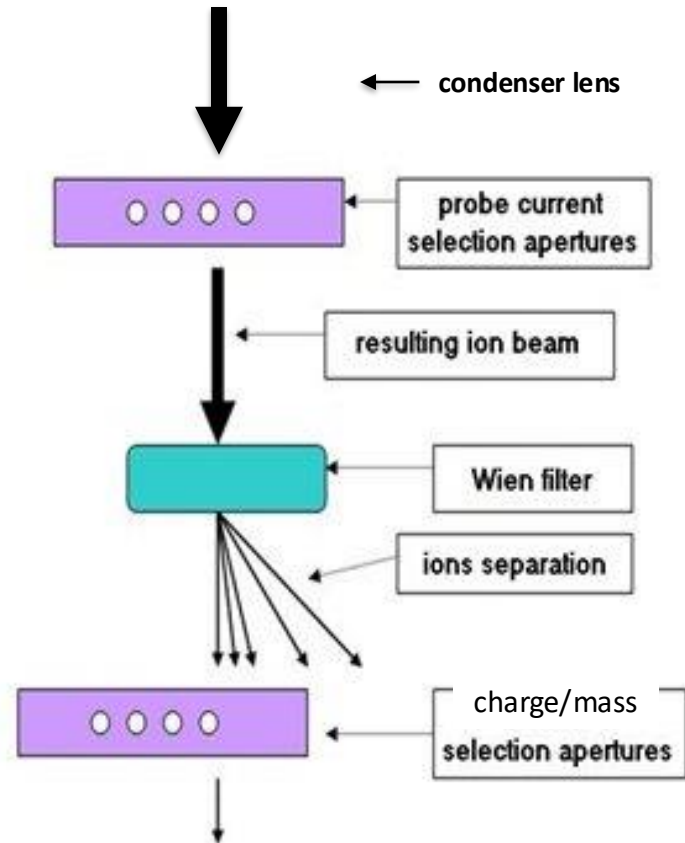


Wien filter

- Ion sources produce a number of ion species corresponding to different:
 - Charge states (eg: Ga^+ , Ga^{2+} ...)
 - Elements in an alloy LMIS (eg: AuGe, AuBeSi, GeMn, PtB)
 - Impurities (eg: residual gases in a plasma source)
 - Isotopes (eg: ^{124}Xe , ^{125}Xe ... ^{136}Xe in a plasma FIB)
- A Wien filter (and aperture/slit) is used to reject unwanted ion species from the beam.

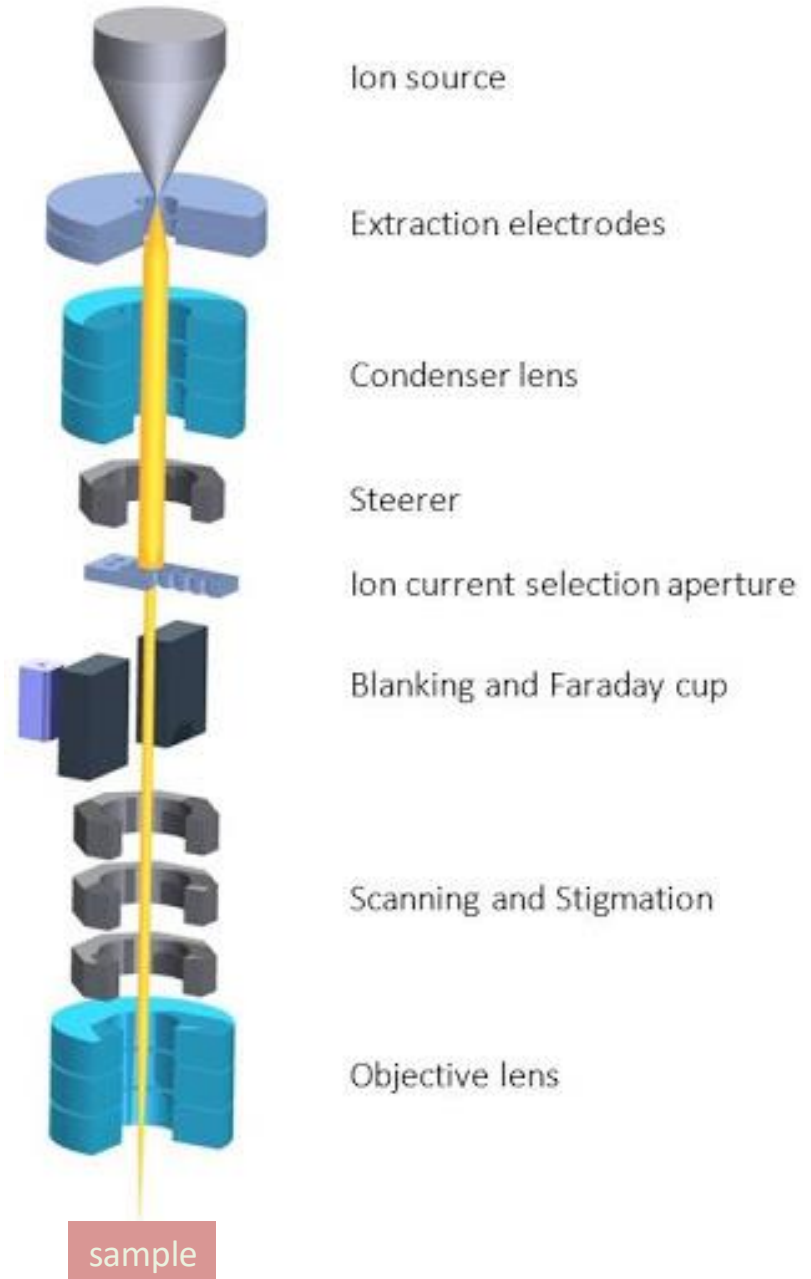


$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$



Focused ion beam (FIB) column

Beam blanker & Faraday cup

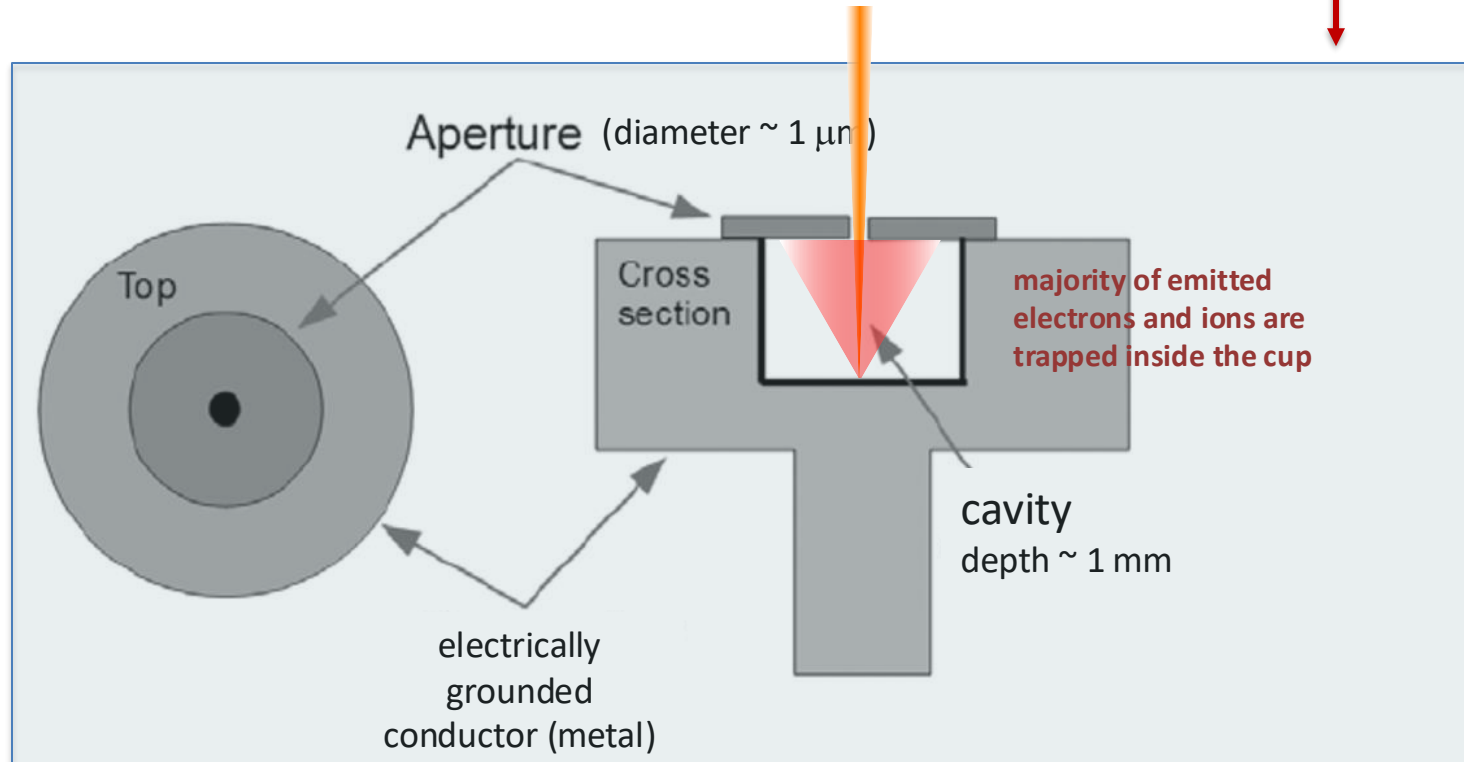


Beam blanker & Faraday cup

Faraday cup can be located:

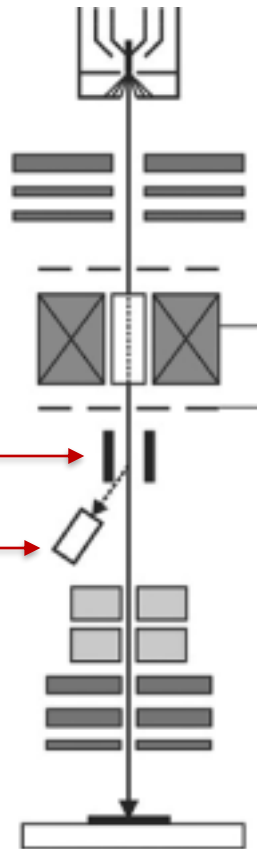
- in the column (as shown here)
- next to the sample

Used in both FIB & SEM instruments.



beam deflection
electrodes

Faraday cup

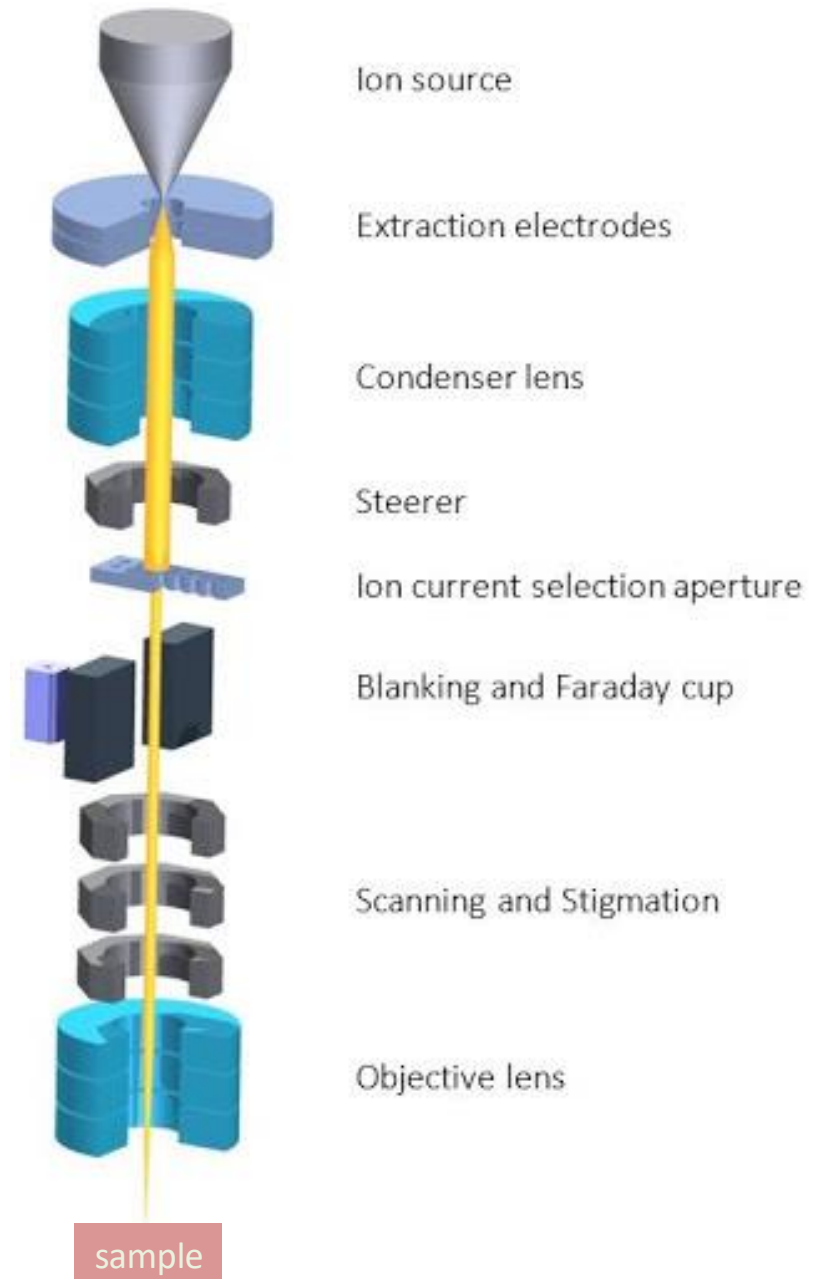


Electrostatic lenses for beam steering and focusing

Condenser lens (and aperture) used to shape beam & reduce current

Electrodes used to correct astigmatism (beam shape) & to scan the ion beam

Objective lens used to focus the beam onto the sample

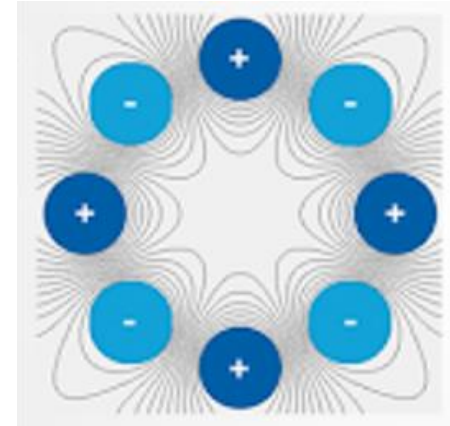


Scanning and astigmatism correction: Multipole ion lenses

quadrupoles used
to guide ions



equipotentials of an
octupole ion lens



Potential field distributions for
common multipoles:

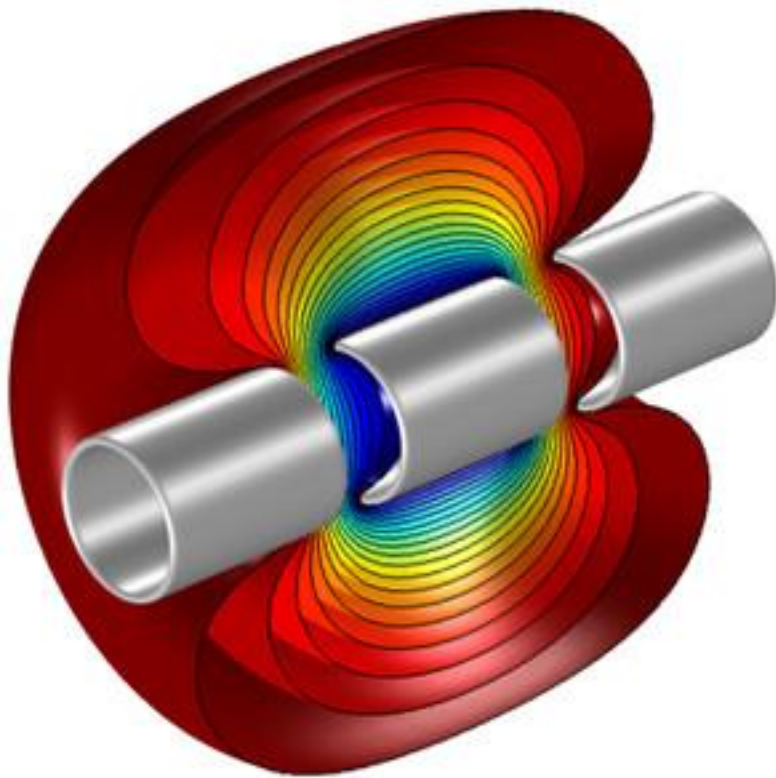
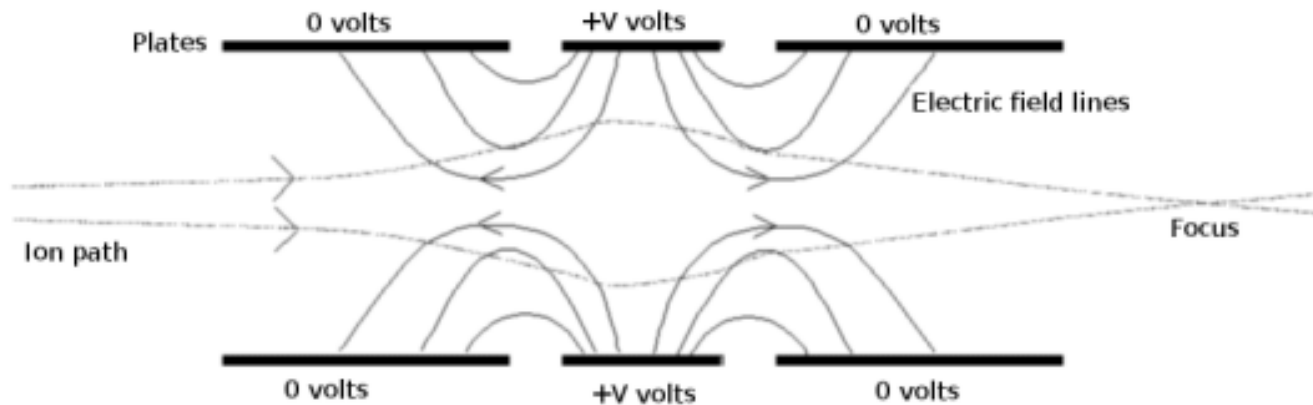
Quadrupole (N=2): $\Phi_2(x,y) = (x^2 + y^2)/(r_0^2)$

Hexapole (N=3): $\Phi_3(x,y) = (x^3 - 3xy^2)/(r_0^3)$

Octopole (N=4): $\Phi_4(x,y) = (x^4 - 6x^2y^2 + y^4)/(r_0^4)$

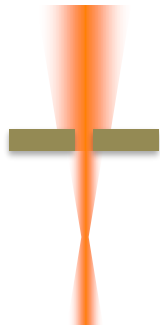
Decapole (N=5): $\Phi_5(x,y) = (x^5 - 10x^3y^2 + 5xy^4)/(r_0^5)$

Einzel electrostatic ion focusing lens



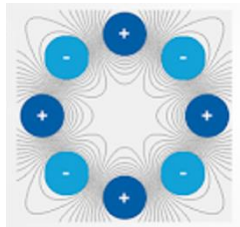
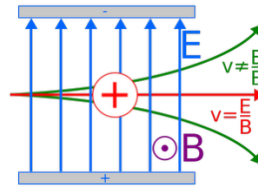
- consists of three or more sets of cylindrical or rectangular tubes in series along an axis
- the lens potential is symmetric
- focuses ions without changing the energy of the beam

FIB column: Summary

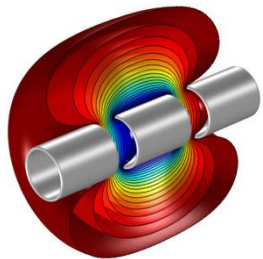


Einzel condenser lens (and an aperture) used to reduce the beam current

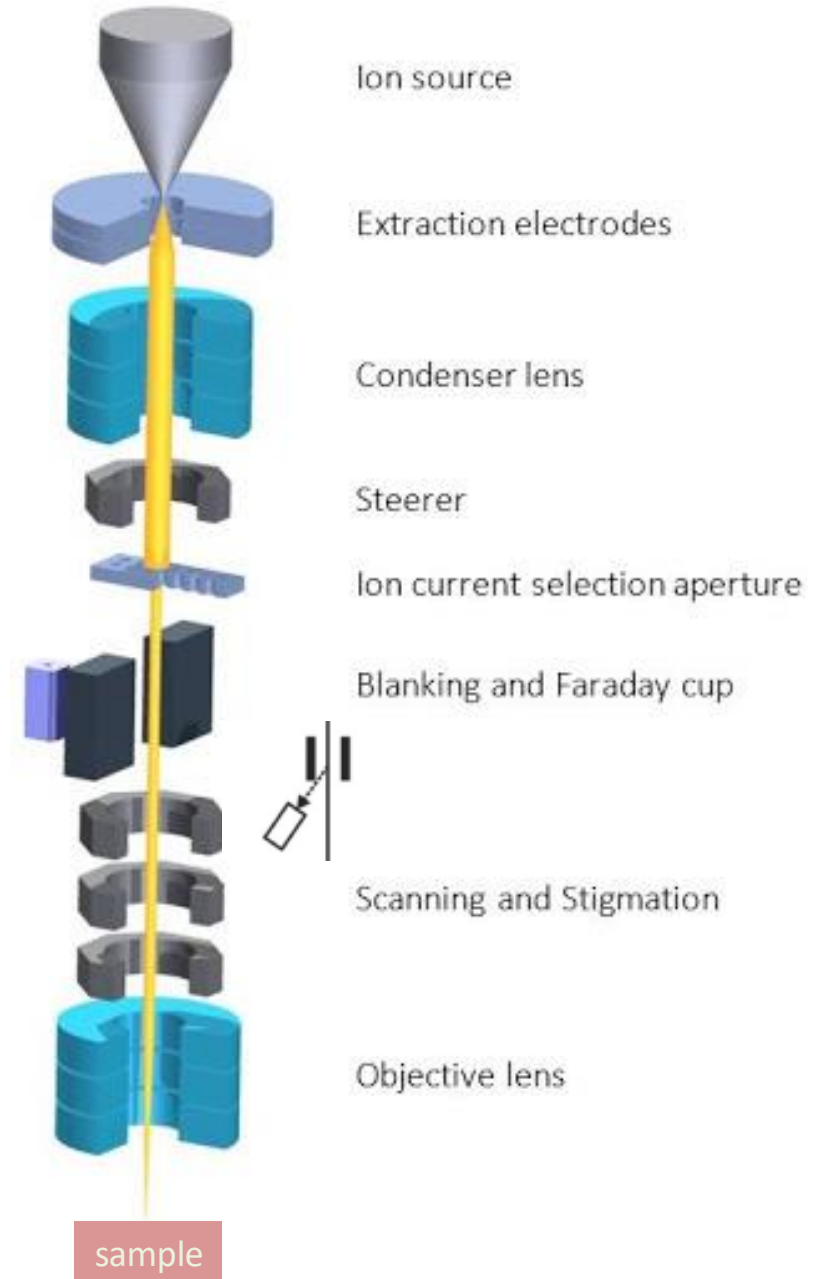
Wien filter (and a slit or aperture) is sometimes used to select the ion species [not shown in this column schematic]



Octupoles used to correct astigmatism (beam shape) & to scan the ion beam



Objective lens used to focus the beam onto the sample (there is a number of different lens types)



Ions vs electrons

electrons

light ions

heavy ions

mass; momentum: $p = mv$ (at a given energy)

velocity: $E = 0.5 mv^2$ (at a given energy)

penetration range (in a sample)

sample interactions & DAMAGE

nanofabrication SPEED (throughput)

ions can remain in the sample – ion “implantation” / “doping”

Focused Ion Beam Microscopy and Micromachining

C.A. Volkert and A.M. Minor, Guest Editors

Abstract

The fairly recent availability of commercial focused ion beam (FIB) microscopes has led to rapid development of their applications for materials science. FIB instruments have both imaging and micromachining capabilities at the nanometer-micrometer scale; thus, a broad range of fundamental studies and technological applications have been enhanced or made possible with FIB technology. This introductory article covers the basic FIB instrument and the fundamentals of ion-solid interactions that lead to the many unique FIB capabilities as well as some of the unwanted artifacts associated with FIB instruments. The four topical articles following this introduction give overviews of specific applications of the FIB in materials science, focusing on its particular strengths as a tool for characterization and transmission electron microscopy sample preparation, as well as its potential for ion beam fabrication and prototyping.

Introduction

The focused ion beam (FIB) microscope has gained widespread use in fundamental materials studies and technological applications over the last several years because it offers both high-resolution imaging and flexible micromachining in a single platform.

The FIB instrument is similar to a scanning electron microscope (SEM), except that the beam that is rastered over the sample is an ion beam rather than an electron beam. Secondary electrons are generated by the interaction of the ion beam with the sample surface and can be used to obtain high-spatial-resolution images. In most commercially available systems, Ga ions are used, and their sputtering action enables precise machining of samples. In conjunction with the gas-injection capabilities on these systems, which enable ion-beam-activated deposition and enhanced etching, a range of sample fabrication schemes are possible.

During the last 25 years, FIB instrumentation has become an important technology for a wide array of materials science applications, from circuit editing and transmission electron microscopy (TEM) sample preparation to microstructural analysis and prototype nanomachining. Most modern FIB instruments supple-

ment the FIB column with an additional SEM column so that the instrument becomes a versatile "dual-beam" platform (FIB-SEM, see Figure 1) for imaging, material removal, and deposition at length scales of a few nanometers to hundreds of microns. The FIB instrument becomes a powerful tool for nanomanipulation and fabrication through the augmentation of an FIB instrument with micromanipulators and gas injection for local chemical vapor deposition (CVD).

The first FIB instruments evolved from advances in field ion microscopes¹ and through the development of high-resolution liquid metal ion sources (LMISs).²⁻⁴ In the 1980s, FIB instruments were embraced by the semiconductor industry as offline equipment for mask or circuit repair. It was not until the 1990s that FIB instruments began to be used in research laboratories, and today there are commercial instruments available from multiple manufacturers.⁵ With the popularity of FIB instruments for TEM sample preparation, microstructural analysis, and nanomachining, dual-beam FIB instruments are becoming a versatile and powerful tool for materials researchers.

This introductory article focuses on the FIB instrument itself and the basic

ion-solid interactions that lead to the various functionalities of FIBs. In the topical articles that follow, the major sub-specialties of FIB research are discussed.

The FIB Instrument

The basic functions of the FIB, namely, imaging and sputtering with an ion beam, require a highly focused beam. A consistent tenet of any focused beam is that the smaller the effective source size, the more current that can be focused to a point. Unlike the broad ion beams generated from plasma sources, high-resolution ion beams are defined by the use of a field ionization source with a small effective source size on the order of 5 nm, therefore enabling the beam to be tightly focused.

The ion source type used in all commercial systems and in the majority of research systems designed with micromachining applications in mind is the liquid-metal ion source (LMIS).^{6,7} Of the existing ion source types, the LMIS provides the brightest and most highly focused beam (when connected to the appropriate optics). There are a number of different types of LMIS sources, the most widely used being a Ga-based blunt needle source. Ga has decided advantages over other LMIS metals such as In, Bi, Sn, and Au because of its combination of low melting temperature (30°C), low volatility, and low vapor pressure. The low melting temperature makes the source easy to design and operate, and because Ga does not react with the material defining the needle (typically W) and evaporation is negligible, Ga-based LMISs are typically more stable than other LMIS metals. During operation, Ga flows from a reservoir to the needle tip (with an end radius of about 10 µm), where it is extracted by field emission. A large negative potential between the needle and an extraction electrode generates an electric field of magnitude 10¹⁰ V/m at the needle tip. The balance between the electrostatic forces and the Ga surface tension wetting the tapered W needle geometry results in the formation of a single Taylor cone at the needle tip. For typical emission currents used in FIB microscopes (~2 µA), a cusp forms at the tip of the Taylor cone with a tip radius of approximately 5 nm.

The simplest and most widely used ion beam columns consist of two lenses (a condenser and objective lens) to define the beam and then focus it on the sample, beam-defining apertures to select the beam diameter and current, deflection plates to raster the beam over the sample surface, stigmation poles to ensure a spherical beam profile, and a high-speed beam blanker to quickly deflect the beam

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Recent advances in focused ion beam nanofabrication for nanostructures and devices: fundamentals and applications

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The past few decades have witnessed growing research interest in developing powerful nanofabrication technologies for three-dimensional (3D) structures and devices to achieve nano-scale and nano-precision manufacturing. Among the various fabrication techniques, focused ion beam (FIB) nanofabrication has been established as a well-suited and promising technique in nearly all fields of nanotechnology for the fabrication of 3D nanostructures and devices because of increasing demands from industry and research. In this article, a series of FIB nanofabrication factors related to the fabrication of 3D nanostructures and devices, including mechanisms, instruments, processes, and typical applications of FIB nanofabrication, are systematically summarized and analyzed in detail. Additionally, current challenges and future development trends of FIB nanofabrication in this field are also given. This work intends to provide guidance for practitioners, researchers, or engineers who wish to learn more about the FIB nanofabrication technology that is driving the revolution in 3D nanostructures and devices.

1. Introduction

Recent years have witnessed rapidly increasing research interest in functional nanostructures and devices with configurations from 0D to 3D. Owing to their unique structural features and excellent physical properties, functional nanostructures and devices are of great significance for a broad range of applications, such as the information industry, optoelectronics, micro/nano electromechanical systems (MEMS/NEMS), biomedical, and micro-energy.¹⁻⁴ The white paper "Nanoscience and Technology: Status and Prospects 2019" predicts that the expected annual global economic impact of nanotechnology closely related to nanostructures and devices, by 2024, will contribute more than \$125 billion US dollars.

For decades, continuous research efforts have been devoted to the development of novel nano-manufacturing technologies that enable the precise fabrication of nanostructures and devices consistent with rational design. To be precise, nanostructures and devices are dependent on nanofabrication tech-

niques to structure matter with nanoscale (refers to the range from 1 to 100 nm) features and nanoscale or even sub-nanometer accuracy. Up to now, a multitude of nano-manufacturing technologies, including optical/electron-beam lithography, nanoimprint lithography (NIL), self-assembly, atomic layer deposition (ALD), chemical mechanical polishing (CMP), laser nanopatterning, 3D printing, and nanotransfer printing (nTP), have demonstrated distinct advantages and revealed enormous potential for the flexible fabrication of nanostructures and devices.^{3,5-7} However, all these nanofabrication technologies have encountered insurmountable problems in the fabrication of arbitrary and high-resolution 3D nanostructures and devices. For instance, lithography techniques have been widely employed for fabricating 2D patterns that can accurately and reproducibly be generated in a specialized layer of material called the resist; nonetheless, those patterns then need to be transferred to another functional layer to achieve 3D fabrication, which is an arduous process.¹ For bottom-up nanofabrication, self-assembly provides a controllable avenue to construct 2D/3D nanostructures with fabrication accuracy at the atomic or molecular level, but it is only applicable to a small number of organic materials and the assembly process cannot be predesigned freely.⁸ To realize atomic or close-to-atomic scale manufacturing, ALD has been widely adopted as a kind of advanced deposition technology with controllable film thickness and excellent shape retention. However, nanostructures and devices with complex configurations cannot be

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Sample test question: Ion beam lens

The operator of a focused ion beam (FIB) microscope increases the magnitude of the voltage applied to a 3 electrode electrostatic Einzel lens. This causes the FIB image to become blurred. Which one of these actions can be used to re-focus the image?

- a) Decrease the working distance.
- b) Increase the ion beam scan speed.
- c) Decrease the ion beam scan speed.
- d) Increase the working distance.
- e) Decrease the ion beam energy.

Sample question: Ion beam column

Consider this schematic of an ion beam column. Why does the focal length (f) of a perfect, symmetric condenser lens affect the spatial resolution of the image?

- a) Because a perfect condenser lens is designed to control the location of the focal plane below the pole piece.
- b) Because f alters the fraction of ions transmitted through the current-limiting aperture.
- c) Because a perfect condenser lens is designed to increase the energy distribution of the ions.
- d) Because a perfect condenser lens is designed to decrease the energy distribution of the ions.

